Strategic Underreporting Under Conditional Access: Evidence from Deep-Sea Mining

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Abstract

The International Seabed Authority (ISA) requires deep-sea mining contractors operating in international waters beyond national jurisdiction to conduct biological surveys of their prospectively-mined tracts, creating a public good as a byproduct of private exploration. However, contractors face a principal-agent problem: they have incentives to underinvest in species detection if discoveries could create obstacles to mining approval. I test for this "taxonomic deflation" using over 160,000 biological samples from the ISA's DeepData database, comparing taxonomic reporting detail in prospectively-mined tracts versus non-contract areas. I show that samples from non-contract areas are significantly more likely to report species-level identification, a pattern that persists after controlling for contractor identity, year, and environmental variables. A counterfactual exercise suggests that tens of thousands of additional observations would include species identifications absent the observed reporting gap. I discuss the ISA's exploration requirement as a second-best solution to the deep sea's public goods problem and consider policy options for mitigating the resulting incentive distortions.

1 Introduction

Can the imprecision of deep-sea mining data be explained by scientific difficulties alone? Biological exploration of the deep sea is genuinely difficult. Depths exceeding four kilometers require specialized remotely operated vehicles, and recovered specimens are frequently damaged in collection or chemically altered by fixation in ways that obscure diagnostic features. Furthermore, the expertise to identify them to species is scarce (Frutos et al. 2022). Much identification must in any case be made from seafloor imagery rather

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than physical specimens, which caps the resolution achievable (Horton et al. 2021), and even intact material often cannot be resolved by morphology alone (Kürzel et al. 2022). Yet the parties responsible for most of this exploration are not entirely impartial: they are mining contractors seeking permits to extract nodules from the seafloor, whose commercial interests do not necessarily align with scientific thoroughness. The Clarion-Clipperton Zone (CCZ), the most active target for deep-sea mining, has yielded a checklist of 5,578 recorded metazoan species (of which only 436 have been formally named), with an estimated richness of 6,200 to 7,600 species (Rabone, Wiethase, et al. 2023). Recent surveys continue to find that most newly collected species are previously undescribed (Jażdżewska et al. 2025). In such conditions, coarse identification could reflect the limits of the discipline—or it could be driven by strategic incentives. A contractor whose sampling uncovers a species that triggers conservation review faces delays, regulatory redesigns, and potential permit denial. I argue that the observed imprecision of deep-sea mining data is partially explained by strategic choices on the part of mining contractors, not by scientific difficulties alone.

This incentive structure is not unique to ocean governance. Wherever a regulator relies on a regulated party to produce the information that determines whether the regulator restricts its activities, the regulated party has reason to underproduce precision. Landowners harboring potentially endangered species have little reason to report or survey them under the Endangered Species Act (Brown and Shogren 1998; Polasky and Doremus 1998); firms required to self-monitor pollutant emissions report values that can diverge from independently measured ones (de Marchi and Hamilton 2006); pharmaceutical sponsors whose trial results determine approval display patterns of selective reporting (Turner et al. 2008). What distinguishes the deep-sea mining setting empirically is that the same firm produces records under both regulatory regimes within the same field campaign, allowing the strategic behavior to be cleanly measured rather than inferred.

Environmental impact assessments are critical for deciding whether, how, and where mining should be permitted. Naturally, those assessments can only be as good as the biological exploration that feeds into them. Unfortunately, no single entity has aligned incentives to fund that exploration: mining contractors gain nothing from species discoveries that do not serve their commercial interests, and third-party explorers do not bear the opportunity cost of delaying mining if they survey slowly. The ISA's operative response has been to make mining contractors do the exploration themselves as a precondition for permit approval. The arrangement uses prospective mining rents as a carrot for biodiversity reporting, but it creates a possible perverse incentive: contractors have reason to underexplore precisely those tracts whose biology bears on their own permit. I ask whether contractors engage in "taxonomic deflation" on those prospectively-mined tracts, by reporting exploration results at lower taxonomic precision relative to the non-contract areas where the same firms also occasionally collect samples. Notably, this latitude is built into the regulatory architecture

itself, as the binding rules compel environmental data collection but leave the *level* of taxonomic resolution to non-binding technical guidance. Deflation thus requires no falsification and no rule-breaking—it can operate on a margin of discretion the ISA has expressly left open.

The ISA requires mining contractors to submit biological data from their exploration areas to the DeepData database. I exploit the fact that the same contractors also occasionally submit data from areas outside their own exploration contracts, where none of their permits is at stake. Most of these records come from the prospectively-mined tracts of *other* contractors, and the rest come from Areas of Particular Environmental Interest (APEIs) and from seabed outside any designated area. I refer to them collectively as non-contract areas. This within-contractor, across-zone comparison holds fixed the inherent difficulties of deep-sea taxonomy and the reporting infrastructure, isolating the effect of the permit incentive on reporting precision.

I begin with a theoretical model in which a contractor jointly chooses two instruments: exploration intensity (how many samples to collect) and taxonomic precision (how specifically to identify the organisms recovered). The two instruments face asymmetric regulatory incentives. More sampling unambiguously helps the contractor by demonstrating survey compliance. Precision cuts the other way: identifying organisms to species level is what risks uncovering a find that triggers conservation review and jeopardizes the permit, although the model allows for other benefits to precision. Outside contract areas, where no permit is at stake, the tension disappears. The model yields two testable predictions: first, that contractors should sample prospectively-mined tracts more intensively than non-contract areas; and second, that they should report those samples with less taxonomic precision.

I test the predictions on over 160,000 biological specimen records from the CCZ, submitted by 12 mining contractors to DeepData. The dependent variable is whether each record carries a species-level identification. I estimate linear probability models that incrementally add controls—a baseline comparison of zone types, then contractor fixed effects (absorbing cross-firm differences in scientific capacity), year fixed effects (absorbing improvements in identification methods over time), and environmental controls matched to each sample's coordinates from Bio-ORACLE rasters. The headline finding is robust across all four specifications: non-contract-area samples are more likely to include a species identification than prospectively-mined-tract samples, and the gap survives increasingly demanding controls. As a robustness exercise, I progressively condition on taxonomy fixed effects from phylum through genus; the gap persists even when I compare organisms classified to the same genus, ruling out compositional differences in sampled taxa as an explanation.

I then exploit a structural feature of the dataset—that a small but consistent share of biological samples are collected by one contractor inside *another* contractor's polygon—to run two complementary robustness

designs. A within-sampler comparison of own-tract vs. guest-tract reporting holds scientific capacity fixed at the firm level, whereas a within-polygon comparison of owner-collected vs. guest-collected samples holds ecology fixed at the tract level. Both designs deliver coefficients consistent with the headline result.

To probe the channel through which deflation operates, I test the addition of cruise-leg, sampling-gear, identifier-name, and laboratory fixed effects to the main specification. Cruise-leg effects absorb roughly 85 percent of the documented gap, which suggests that much of the deflation is decided at the level of the expedition: which cruises go where, with what personnel and equipment. This aligns naturally with the level of contracting, as mining contractors will typically arrange to fund exploration at the cruise level. Sampling gear does not explain any of the observed gap—if anything, accounting for gear types amplifies the result, since somewhat more informative equipment is used on prospectively mined tracts than elsewhere. The volume of samples does not explain it either. Identifier-name effects also absorb the gap among the records that name an identifier, and the gap is likewise absent within the laboratories that perform the identifications. However, almost all taxonomists work for a single firm, so these effects cannot separate the individual from the team to which a contractor assigns its samples. Taken together, these results suggest that the distortion is a matter of how a contractor organizes its survey work, rather than a technical limit of the equipment or a choice made by the individual naming the specimen.

Next I run a counterfactual exercise that simulates species reporting as if every observation were reported with non-contract-area-level precision, holding exploration intensity and all other covariates fixed. The exercise decomposes the missing species-level identifications in the dataset into two parts: a portion attributable to strategic incentives (the deflation wedge) and a portion attributable to the genuine limits of deep-sea taxonomy. Even under the most conservative coefficient, strategic incentives account for roughly thirteen percent of missing species identifications—on the order of 15,000 specimens under the conservative bound and as many as 43,000 under the largest coefficient, all of which would carry a species-level identification absent taxonomic deflation.

A companion exercise turns to the underexploration outside contract areas. APEIs, the conservation blocks the ISA designed as a representative biodiversity benchmark, are surveyed more than 600 times less densely than the tracts contractors intend to mine; had they been sampled at mined-tract density, the few hundred specimens they have produced would instead run to the hundreds of thousands, carrying on the order of 81,000 additional species-level identifications. While these counterfactual thought experiments are large extrapolations, they suggest strategic incentives play a large role in explaining the lack of biodiversity data from the deep sea.

I close with policy options, organized around the two wedges the model identifies—though, as the model also shows, on a contract area neither instrument is confined to a single margin. Section 2 presents the institutional background, Section 3 the model, Section 4 the empirical methods, Section 5 the results (with the mechanism analyses), Section 6 the interpretation, Section 7 policy responses, and Section 8 concludes.

My analysis draws on three strands of the economics literature. The first is the literature on strategic manipulation of self-reported environmental data. Firms' self-reported environmental data can diverge from independently measured values (de Marchi and Hamilton 2006), and third-party auditors can be compromised when their compensation depends on the regulated firm (Duflo et al. 2013). Regulated agents respond dynamically to enforcement, adapting their evasion strategies as they learn the inspection regime (Gonzalez-Lira and Mobarak 2021), and regulatory pressure can paradoxically reduce the quantity and quality of voluntary disclosures (Zakota 2025). In some cases agents manipulate the monitoring process itself rather than the data it produces.[1] Mandatory disclosure requirements can meaningfully improve environmental outcomes (Bennear and Olmstead 2008; Doshi et al. 2013), though firms also selectively disclose favorable information while withholding unfavorable results (Lyon and Maxwell 2011; Kim and Lyon 2011).[2] The form of misreporting I document operates through a different margin: contractors do not fabricate coarse identifications but rationally forgo the precise ones.

The second strand concerns perverse incentives in species-protection regulation. Brown and Shogren (1998) and Langpap et al. (2018) survey the economics of the Endangered Species Act (ESA), emphasizing the tension between conservation benefits that accrue nationally and costs that fall on private actors harboring species. The tension generates strategic responses: landowners near known endangered-species colonies preemptively destroy potential habitat to avoid triggering ESA protections (Lueck and Michael 2003), and ESA policy gives landowners little incentive to cooperate with species-information collection on their land (Polasky and Doremus 1998). Institutional design can mitigate these incentives: safe-harbor agreements reduce preemptive habitat destruction (Byl 2019). The mechanism in my setting is structurally similar—an agent with a valuable asset manipulates the conditions that would trigger conservation oversight—but the manipulation is informational rather than physical.

The third strand is the literature on regulatory enforcement and compliance under asymmetric information (Harrington 1988; Heyes and Rickman 1999). A consistent finding is that enforcement delegated to parties with a financial stake in the outcome tends to break down: drivers in Mexico City bribe emissions inspectors

[1] Mu et al. (2026) show that local governments strategically shut down air-pollution monitors on days when air quality is expected to deteriorate.

[2] Kim and Lyon (2011) find that firms in the DOE's Voluntary Greenhouse Gas Registry reported emissions reductions even as their actual emissions increased.

rather than repair their cars (Oliva 2015), and competition between Texas testing firms actually increases falsification, because drivers can shop around for a lenient inspector (Lugthart 2026). Notably, those settings still involve a nominally independent intermediary, whereas the ISA's does not: the firm whose permit is at stake is itself producing the environmental record on which that permit will be judged. The ISA's exploration requirement uses private rents as a carrot for public-good provision, and the taxonomic deflation I document is precisely the kind of shirking on uncontracted margins that such arrangements invite.

Across these strands, I develop three conceptual contributions. First, I distinguish *strategic imprecision*, the rational underproduction of information that the agent is required to produce, from the more familiar problems of falsification, evasion, and selective disclosure. Where misreporting requires the agent to assert a false fact, strategic imprecision requires only that the agent stop short of a true one. That is, the channel I document does not require any deception—it requires only an institutional setting in which silence is cheaper than precision. I also coin a specific case of strategic imprecision, *taxonomic deflation*, which is the inverse of taxonomic inflation (Isaac et al. 2004). Second, I provide a tractable model in which exploration intensity and reporting precision are chosen jointly, showing that the prospect of a mining permit is on its own enough to raise sampling and depress precision at the same time and that the resulting welfare loss has two distinct sources. Third, I locate the *level* at which strategic imprecision operates. The mechanisms behind the taxonomic precision gap are choices about how to organize exploratory activity (e.g., which expedition goes where, with which team, and where the samples are sent subsequently), whereas the gap does not appear within the work of a given identifier or laboratory. The policy-relevant unit is therefore the firm's allocation of identification effort, with implications for any setting in which the regulated firm retains discretion over who analyzes the information used to evaluate it.

2 Background

2.1 The Deep Sea as a Commons

The seabed beyond national jurisdiction and its mineral resources are designated "the common heritage of mankind" under the United Nations Convention on the Law of the Sea (UNCLOS). Those minerals include cobalt, nickel, copper, and manganese. However, the same seabed is also home to thousands of species, most of which have never been described, and the ISA is required to protect them from the harmful effects of mining (International Seabed Authority 2023). The minerals have immediate commercial value and could prove useful for decarbonization, but deep-seabed mining is widely expected to destroy valuable biodiversity in the process (Niner et al. 2018; Levin et al. 2020). The concern is not speculative. Jones

et al. (2025) revisited experimental mining tracks in the CCZ forty-four years after disturbance and found that large sessile fauna had barely recovered; an industrial-scale mining trial documented a 37% reduction in macrofaunal abundance and a 32% reduction in species richness in directly impacted areas (Stewart et al. 2026). The baseline against which those losses must be measured is barely characterized at present (Kennedy and Rotjan 2023): less than 0.001% of the deep seabed has been visually observed (Bell et al. 2025), and Rabone, Wiethase, et al. (2023) place total richness in the CCZ at 6,200 to 7,600 species, of which 5,578 have been recorded and only 436 formally named. Bonifácio et al. (2021) document substantial morphological and genetic diversity among deep-sea scale-worms (Polynoidae); Jażdżewska et al. (2025) catalog the CCZ's Amphipoda and find that most discovered species were previously unknown. So before any decision about mining can be made on a scientifically grounded basis, biological exploration has to happen. Yet exploration has historically fallen short of describing the rich variety of life in the deep sea. Species discovery and identification are rarely funded (Glover et al. 2018; Britz et al. 2020)—a shortfall that Glover et al. (2018) calls the "historic failure of taxonomy on the high seas."

2.2 The ISA's Exploration Requirement

The ISA manages the deep seabed and has issued exploration contracts entitling country-backed contractors to survey a specific tract (Blanchard et al. 2023). The ISA officially terms these "exploration areas." Because what drives the incentives I study is that a single contractor holds and expects to mine them, I call them *contract areas* throughout—using *prospectively-mined tracts* interchangeably where the mining prospect is the salient feature—to contrast them with the areas where the same contractor has no permit at stake. The eventual goal of an exploration contract is conversion to an exploitation contract, but the present requirement is that the firm intending to mine an area must first collect environmental and biological data about it. The straightforward justification is that the data are inputs to the ISA's decision over whether mining should be allowed in a particular location. Economically, the requirement forces prospective miners to produce a public good (non-rival, non-excludable information about the marine genetic resources of the deep sea) that the ISA needs in order to balance mining against conservation (Jones et al. 2019).

A central deliberation over whether deep-sea mining should be permitted at all turns on undiscovered species and the precautionary principle (Jaeckel 2017). If firms must fund exploration before exploitation, mining cannot destroy a valuable ecosystem or genetic resource before scientists can characterize it. Deep-sea organisms have become increasingly prominent in marine bioprospecting, with growing numbers of patent filings tied to deep-sea genetic material (Zhivkoplias et al. 2024; Krusberg et al. 2024). If the seabed might harbor valuable biological resources, there is quasi-option value to delaying mining in favor of exploration

(Arrow and Fisher 1974; Henry 1974), particularly given evidence that mining impacts persist for decades and that recovery is at best partial (Jones et al. 2025). The catch is that no one outside the ISA has aligned incentives to fund that exploration on the timeline mining requires. Companies that might eventually benefit from deep-sea genetic discoveries cannot secure exclusive rights to the organisms themselves or to the places where they are found, so they have little reason to fund broad exploration now. With that in mind, the ISA's exploration requirement is a clever workaround: it forces the contractor to bear both the cost of exploration and the opportunity cost of delaying mining, in exchange for the right to mine eventually. The immediate alternatives, such as granting the contractor exclusive rights over biological resources as well as minerals, or pausing mining indefinitely until a scientific body deems the deep sea "sufficiently explored", are legally infeasible and create their own tensions between exploration and conservation.

In practice, the ISA's requirements for exploration leave a lot under the contractor's discretion. The binding instruments are the ISA's exploration regulations and the standard contract clauses they incorporate: each contractor "shall" gather environmental baseline data, establish baselines, run a monitoring program, report the results annually, and its application must describe a program to assess the "impact on biodiversity" of the proposed activities (International Seabed Authority 2013). However, the binding regulations stop there. They never enumerate which taxa must be characterized, never specify a taxonomic level, and route the question of *how* to conduct biological classification to recommendations issued by the Legal and Technical Commission—guidance that contractors must take into account and observe only "as far as reasonably practicable" (International Seabed Authority 2013). Those recommendations, which are the technical rules contractors actually follow, are written almost entirely in the register of "should" rather than "shall" (International Seabed Authority 2023). They explicitly endorse "taxonomy by numbers" (numbered morphospecies backed by voucher specimens) as "a good basis for baseline studies." They treat image-based megafaunal surveys as resolving only organisms larger than two centimeters, and as requiring physical specimens to confirm any species identification. They even allow a contractor to petition for exemption from any recommendation it finds infeasible (International Seabed Authority 2023). An ISA workshop on taxonomic standardization reached the same conclusion: to ensure high-quality taxonomic outputs, "standards must be set and enforced" (International Seabed Authority 2020). The upshot is that a contractor can satisfy every binding obligation—survey the tract, archive its specimens, file its annual reports—while reporting what it recovers only to genus or family, if not coarser. That is, species-level identification is encouraged, but not required.

2.3 Taxonomic Deflation and the Comparison Strategy

As discussed, the ISA's solution to underexploration comes with a possible downside: it can incentivize the contractor to identify what it recovers only coarsely. Because the contractor's returns come only from eventually mining the tract, it has strong reason to avoid discovering anything that might lead to its permit being denied, such as rare species, novel taxa, or charismatic fauna. Species-level identification also matters for conservation status, and for the regulatory review and delay that such status can invite (Delić et al. 2017). This incentive could be partially addressed by mandating high standards for the quality and thoroughness of reporting, by requiring third-party explorers, or by imposing other procedural safeguards. But the incentive can still operate through uncontracted margins: the contractor can hire different teams of scientists, use different equipment, or shade categorization decisions when the underlying classification is uncertain. In particular, the contractor can attempt *taxonomic deflation*—the flip side of "taxonomic inflation," in which subpopulations are elevated to independent species (Isaac et al. 2004). Taxonomic inflation has been used in service of conservation, by making subpopulations eligible for endangered species protection (Ghezelayagh et al. 2025). Taxonomic deflation can do the opposite: it undermines conservation by failing to identify distinct species that should be considered separately for protection. Both manipulations exploit the same underlying fact that "species" can be hard to define precisely, and that difficulty leaves room for parties with their own goals to push the boundary either way (George and Mayden 2005). The species is nonetheless the fundamental unit of biodiversity, so "deflation" to the phylum or class necessarily understates a substantial amount of biodiversity (Claridge et al. 1997).

All biological and environmental samples collected under an exploration contract must be reported to the ISA in a prescribed annual-report format (International Seabed Authority 2015) and are eventually made public through the DeepData database. That format standardizes the *fields* a contractor populates, but it imposes no floor on the taxonomic *resolution* those fields must reach. The quality of the resulting reports varies considerably. Rabone, Horton, et al. (2023) document persistent issues, including duplication of entries, inconsistent or absent identifiers, and substantial data-quality problems in the taxonomy variables. Even absent strategic underreporting, deep-sea biological data are often taxonomically coarse (Horton et al. 2021; Kennedy et al. 2019), and the ISA has been markedly less transparent than other international resource managers (Ardron 2018). Just 28,565 of the 160,967 CCZ records in DeepData (18%) carry a species-level identification at all. Consequently, pointing out the absolute lack of taxonomic information in DeepData is not enough: the data from prospectively-mined tracts has to be compared against an appropriate counterfactual.

That counterfactual is the same contractors' submissions from outside their own tracts—mostly the

prospectively-mined tracts of *other* contractors, plus APEIs and areas outside any designated polygon—which I refer to collectively as non-contract areas. What unites them is not that the ground is unclaimed, but that *this* contractor's permit does not turn on what is found there. Every sample in DeepData, including those from non-contract areas, is submitted by a mining contractor as part of an exploration program. The comparison is therefore within the same reporting entities, holding fixed the inherent difficulties of deep-sea exploration, the DeepData reporting infrastructure, and each contractor's scientific methodology and institutional partnerships. If the same contractors' data show systematically less taxonomic detail in contract areas than in non-contract areas, that pattern is consistent with taxonomic deflation. Figure 1 shows the administrative boundaries of the CCZ, with contractor tracts (colored) and APEIs (gray) labeled. The spatial distribution of biological samples in the DeepData database is shown in Appendix Figure A2.

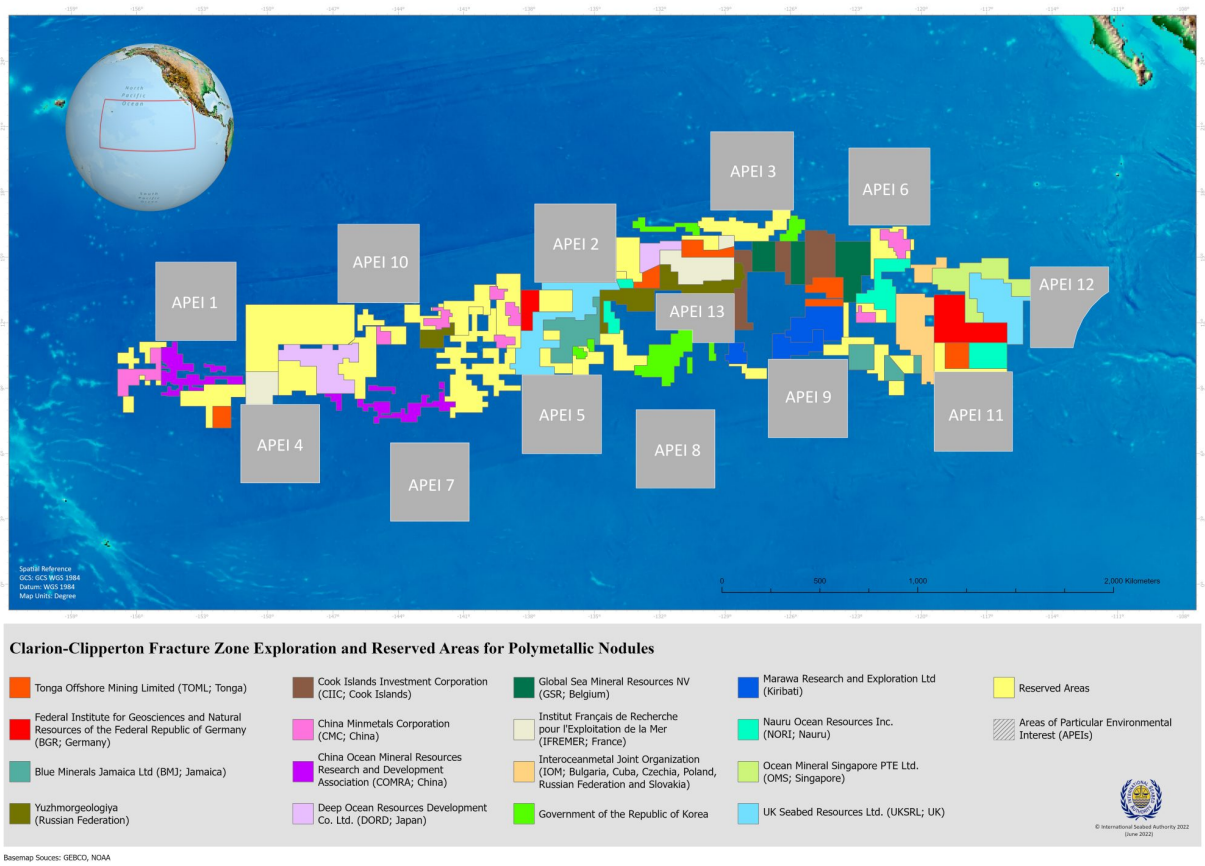


Figure 1: Contractor exploration areas (colored) and Areas of Particular Environmental Interest (gray) in the Clarion-Clipperton Zone. Source: International Seabed Authority.

3 Model

Before turning to the empirics, I develop a model that clarifies the mechanism behind the empirical strategy. The model shows that the observed pattern of taxonomic deflation follows from rational contractor behavior under the ISA's regulatory structure. Two predictions emerge, each corresponding directly to one set of empirical findings below: contractors sample prospectively-mined tracts more intensively than non-contract areas, yet report those samples with less taxonomic precision.

The core idea is that exploration intensity and taxonomic precision are distinct instruments facing asymmetric regulatory incentives. More sampling unambiguously raises a contractor's odds of regulatory approval by demonstrating survey compliance. Precision cuts the other way: carrying an organism to species level can produce a finding, such as a rare or valued species, that triggers conservation review and jeopardizes the mining permit. Precision may still be worth something to the contractor through other channels, such as scientific returns or some regulatory minimums, of course. Outside contract areas, where no permit is at stake, only the scientific return operates, and contractors have no reason to withhold precise identifications.

3.1 Setup

Consider a mining contractor that jointly chooses two instruments: exploration intensity $e \geq 0$, capturing the volume of biological sampling it conducts, and taxonomic precision $\tau \in [0, 1]$, capturing how specifically it identifies the organisms recovered. In practice the activity itself is likely contracted out to scientific partners, so these choices are stand-ins for the contract terms and funding dedicated to this endeavor. Although any single specimen is identified to a discrete taxonomic rank, treating τ as continuous is natural at the level of a survey: it represents the *share* of recovered organisms the contractor carries to species level, which varies smoothly with the identification effort it allocates. The two instruments are conceptually and empirically distinct—a contractor can collect many samples while reporting all observations only at the phylum level, or conduct a small but meticulous survey. The DeepData database encodes exactly this distinction, as it records both the number of biological samples submitted and the taxonomic level at which each is identified.

The regulator observes both instruments and makes an approval decision. I write the probability of approval as a product of two forces:

$$\pi(e, \tau) = \underbrace{A(e)}_{\text{compliance}} \cdot \underbrace{S(\tau)}_{\text{survival}} \tag{1}$$

$A(e)$ is the probability that the survey satisfies the Authority's requirements, and it rises with sampling:

this is the compliance benefit of exploration. $S(\tau)$ is the probability that nothing the contractor identifies blocks the permit, and it falls with precision: this is the discovery hazard. The product form implicitly assumes that the two are independent hurdles—the survey must both satisfy the regulator and turn up nothing disqualifying.

I assume each survey has some value to the contractor separate from the permit, although that value may be arbitrarily small. Let that value be $G(e) + H(\tau)$, where G is the value of a survey's *coverage* and H the value of its *resolution*, each already net of what it costs to produce. In this model, the contractor pushes each instrument to the point where that instrument's own marginal benefit meets its own marginal cost, and does so in each zone separately. This is a natural description of the setting, as sampling happens at sea and identification happens onshore, months later and usually by different people.

Suppose the mining permit is worth $V > 0$. Letting $M = 1$ denote a prospectively-mined tract, the contractor's expected payoff is

$$\Pi(e, \tau; M) = \mathbb{1}\{M\} \cdot A(e)\, S(\tau) \cdot V \;+\; G(e) + H(\tau). \tag{2}$$

The two zones share one objective and differ only by the indicator on the permit term, so setting $V = 0$ returns the contract problem exactly to the non-contract problem. I impose three restrictions:

(A1) $A' > 0, \quad A \in [0, 1)$

(A2) $S' < 0, \quad S \in (0, 1], \quad S'$ bounded

(A3) G, H strictly concave, continuously differentiable, with interior peaks, and $G'(0) = H'(0) = +\infty$.

(A1) assumes that more sampling leads to a higher probability of compliance, at a rate that is everywhere strictly positive.[3] (A2) says that precision raises discovery risk, that no level of precision makes denial certain, and that the risk a marginal identification adds is finite.[4] (A3) says coverage and resolution each pay off with diminishing returns, are eventually not worth what they cost, and that the first unit of each

[3] The strictness matters: a contractor facing a locally flat compliance return at the science optimum would have no reason to sample past it, and Prediction 1 would survive only in weak form.

[4] Bounding S', together with the Inada conditions in (A3), is what keeps the contractor's own optimum interior. As precision approaches zero, the marginal scientific return to resolution grows without bound while the marginal hazard stays finite, so the contractor never abandons identification altogether.

is always worth buying.[5] These minimal assumptions do not impose a specific functional form, but just restrict the *direction* of each force. I do not assume the contractor's objective is concave on a contract area—in general it is not, since a discovery hazard is naturally convex—nor that its optimum is unique.[6]

However, for concreteness and to produce closed forms in later sections, consider the following:

$$A(e) = 1 - e^{-\alpha e}, \qquad S(\tau) = e^{-\phi\tau}, \qquad G(e) = b_e\sqrt{e} - \kappa e^2, \qquad H(\tau) = b_\tau\sqrt{\tau} - \lambda\tau^2,$$

with $\alpha, b_e, b_\tau, \kappa, \lambda > 0$ and $b_\tau < 4\lambda$, which places the resolution peak inside $[0, 1]$. Here α is the rate at which sampling buys compliance, b_e and b_τ are the scientific returns to coverage and resolution, κ and λ are their costs, and $\phi > 0$ is the regulatory bite of a species discovery (the rate at which precision erodes the odds of approval). The cost parameters sit inside the science functions, since G and H are what coverage and resolution are worth net of what they cost. The central predictions hold without these specific functional forms, however.

3.2 Optimal Choices in Non-Contract Areas

In non-contract areas there is no permit at stake, and the contractor faces no regulatory penalty for precise identification. Its problem collapses to

$$\Pi_A(e, \tau) = G(e) + H(\tau), \tag{3}$$

which is additively separable and strictly concave in each argument, so it has a unique solution obtained by setting each instrument at the peak of its own return: $G'(e_A) = 0$ and $H'(\tau_A) = 0$. Off contract, in other words, the contractor runs the survey that its scientific incentives alone would justify. Then the two instruments decouple completely: e_A and τ_A are constants, neither depending on the other, and neither depending on anything about the permit. Under the illustrative forms,

$$e_A = \left(\tfrac{b_e}{4\kappa}\right)^{2/3}, \qquad \tau_A = \left(\tfrac{b_\tau}{4\lambda}\right)^{2/3}. \tag{4}$$

[5] Differentiability is required on the whole interior rather than only at the peaks because the welfare wedges of Section 3.4 are evaluated at the contractor's own optimum—which Predictions 1 and 2 place strictly away from those peaks.

[6] A maximizer exists: $A(e)S(\tau)V \leq V$ is bounded while (A3) sends $G(e) \to -\infty$ at least linearly, so the objective is coercive in e and the problem may be restricted to a compact set on which it is continuous.

3.3 The Central Predictions

Two facts about (2) drive both predictions. First, on a contract area sampling and precision are substitutes: the more the contractor has already sampled, the less an additional identification is worth to it. Formally, for $e' \geq e$ and $\tau' \leq \tau$,

$$\Pi(e', \tau') + \Pi(e, \tau) - \Pi(e', \tau) - \Pi(e, \tau') \;=\; \mathbb{1}\{M\}\, V \left[A(e') - A(e)\right]\left[S(\tau') - S(\tau)\right] \;\geq\; 0.$$

Second, the permit raises the value of sampling and lowers the value of precision: $\Pi(\cdot\,; 1) - \Pi(\cdot\,; 0) = A(e)S(\tau)V$ is increasing in e by (A1) and decreasing in τ by (A2). Both facts use only the monotonicity of A and S, so they do not depend on the specific functional forms, only on the minimal assumptions imposed above.

These are the conditions under which the theory of monotone comparative statics applies, provided the precision dimension is read in reverse, with the contractor choosing *coarseness* rather than precision (Topkis 1978; Milgrom and Shannon 1994).[7] Read that way, the contractor's two choices are complements, since the more it has sampled the more it wants to deflate, and the permit raises the return to both at once. The theorem then says that raising the parameter, here the contract indicator itself, raises the contractor's optimal choice of each: it samples more and identifies more coarsely. Because the non-contract optimum is a single point, this holds for *every* optimal choice the contractor might make on a contract area, not only the extreme ones.[8] Hence $e_M \geq e_A$ and $\tau_M \leq \tau_A$.

Strictness follows from evaluating Π_M at the benchmark, which (A3) places in the interior. Since $S(\tau) > 0$ for every τ by (A2), we get

$$\left.\frac{\partial \Pi_M}{\partial e}\right|_{e_A} = \underbrace{A'(e_A)\, S(\tau)\, V}_{>\,0} + \underbrace{G'(e_A)}_{=\,0} \;>\; 0 \quad \text{for any } \tau, \tag{5}$$

so e_A is not a critical point of the contract-area problem and no optimum sits there. That gives $e_M > e_A > 0$, so $A(e_M) > A(0) \geq 0$ strictly (A being strictly increasing) and therefore

[7] Formally, order the choice set by $(e, -\tau)$, which reverses the order on the second dimension and leaves the choice set a lattice. Under that order the two displayed facts are in fact the supermodularity of Π and increasing differences in M, which are the theorem's hypotheses.

[8] The conclusion of the theorem is that the contract-side optimal set dominates the non-contract set in the strong set order: for any y in the former and x in the latter, the componentwise minimum of y and x lies in the latter. When the latter is the singleton $\{(e_A, -\tau_A)\}$, that minimum must equal $(e_A, -\tau_A)$ itself, which says precisely $y \geq (e_A, -\tau_A)$—that is, $e_M \geq e_A$ and $\tau_M \leq \tau_A$.

$$\left.\frac{\partial \Pi_M}{\partial \tau}\right|_{\tau_A} = \underbrace{A(e_M)\, S'(\tau_A)\, V}_{<0} + \underbrace{H'(\tau_A)}_{=0} \;<\; 0. \tag{6}$$

Collecting these results gives two stark empirical predictions:

Prediction 1: More Exploration on Prospectively-Mined Tracts. *For any $V > 0$, the contractor explores a prospectively-mined tract more intensively than a non-contract area: $e_M > e_A$.* This result follows straightforwardly from the fact that sampling is a compliance requirement, which is only worth something when a permit is at stake. Outside of contract areas the contractor samples only as much as its private scientific returns warrant; on contract it samples more, at any positive permit value, because (5) is strictly positive whenever $V > 0$.

Prediction 2: Less Precision on Prospectively-Mined Tracts. *For any $V > 0$, the contractor reports a prospectively-mined tract less precisely than a non-contract area: $\tau_M < \tau_A$.* Outside of contract areas, taxonomic precision carries no regulatory risk and the contractor identifies as warranted by its private scientific returns. Within contract areas, however, additional identification carries the risk of a finding that blocks the permit, so the contractor stops short: by (6), it is already better off pulling back from the precision that science alone would warrant.

The two predictions are not strictly independent. While sampling and precision are independent outside of a contract area, they are substitutes within one: $\partial^2 \Pi / \partial e\, \partial \tau = \mathbb{1}\{M\}\, V A'(e) S'(\tau)$ is negative when $M = 1$ and zero when $M = 0$. The hazard's bite is proportional to $A(e)$, the compliance value the contractor's sampling has earned. Consequently, the more a contractor has surveyed, the more an adverse discovery costs it, and the stronger its incentive to report coarsely.

3.4 Further Implications

The model generates additional comparative statics that, while not directly tested here, sharpen intuition about how the deflation mechanism operates.

Asset value and deflation intensity. More valuable tracts should draw more sampling and coarser identification, all else equal. e_M is nondecreasing and τ_M nonincreasing in V, by the same argument as the predictions with V in place of the contract indicator, since raising V raises the marginal value of sampling ($\partial^2 \Pi / \partial e\, \partial V = A' S > 0$) and lowers that of precision ($\partial^2 \Pi / \partial \tau\, \partial V = A S' \leq 0$). More commercially valuable deposits, those with greater nodule density or surveyed during periods of high commodity prices, should produce larger sample counts but coarser identifications within the same contractor. Testing this directly

would require estimates of nodule abundance or value, which are not systematically available. However, even with those estimates, contractors may differ in their non-permit returns and costs, so the prediction is only conditional.

Regulatory bite. The deflation gap need not widen as the consequences of a species discovery grow more severe. Because a larger ϕ also deters sampling, and because sampling and precision are substitutes on a contract area, less sampling pushes precision back up. As written, the model does not sign the net effect, though a larger regulatory bite unambiguously lowers the *marginal* return to sampling. In any case, testing the effect of regulatory bite directly would require variation in regulatory stringency, which is not available.

These behavioral implications have direct welfare consequences. Consider a social planner who values a precisely-identified species at $s > 0$, capturing its option value for conservation, ecosystem services, and future research. The social planner does not consider a discovery as a loss, since blocking mining redirects the tract to protection rather than destroying surplus. Their objective function is the following:

$$W(e, \tau) = A(e)\, V + G(e) + H(\tau) + s \cdot e \cdot \tau. \tag{7}$$

The scientific term enters unweighted by the approval probability: the knowledge from a precisely-identified sample accrues to society whether or not the permit is ultimately granted, unlike the contractor's permit payoff which is contingent on approval. The discovery hazard, meanwhile, enters the contractor's objective but not the planner's—society is not harmed by learning that a tract holds a rare species.[9] Evaluating the planner's marginal valuations at the contractor's own optimum on a prospectively-mined tract gives two wedges, both strictly positive:

$$\left.\frac{\partial W}{\partial \tau}\right|_{(e_M, \tau_M)} = \underbrace{-A(e_M) S'(\tau_M) V}_{\text{hazard the planner does not face}} + \underbrace{s\, e_M}_{\text{unpriced information}}, \tag{8}$$

$$\left.\frac{\partial W}{\partial e}\right|_{(e_M, \tau_M)} = \underbrace{A'(e_M) V \big[1 - S(\tau_M)\big]}_{\text{hazard also deters sampling}} + \underbrace{s\, \tau_M}_{\text{unpriced information}}. \tag{9}$$

The planner wants more of *both* sampling and precision than the contractor supplies. The precision wedge rests on weaker assumptions than the exploration wedge. Prediction 2 puts τ_M strictly below τ_A, and H

[9] Two features of (7) are normalizations here. Society is taken to value an approved tract at the same V the contractor does, and the compliance term is retained unweighted by S, which treats a permit blocked by a discovery as costless to society but a permit blocked by non-compliance as a loss. Weakening either changes the magnitude of the wedges, but not the sign of the precision wedge.

is strictly concave with its peak at τ_A, so $H'(\tau_M) > 0$ and $\partial W/\partial \tau = H'(\tau_M) + s\, e_M > 0$ follows without any appeal to the contractor's first-order condition. A different kind of exploration wedge is present in non-contract areas too. There society's problem is $W_A(e, \tau) = G(e) + H(\tau) + s\, e\, \tau$, with no permit to grant, so evaluating at the contractor's own choice gives $\partial W_A/\partial e|_{e_A} = s\, \tau_A$, the unpriced information alone. Neither ϕ nor V appears, meaning off contract the shortfall in sampling answers only to instruments that reward discovery.

Two distinct sources of divergence appear, with different remedies. The first is the permit risk, ϕ, which the contractor fears and the planner does not. Reducing it—through safe-harbor provisions, sealed reporting, or independent review of contested identifications—can reduce the first term of each wedge (at costs discussed later). The second is the scientific value s, which society enjoys and the contractor does not. Making the contractor internalize it—through prizes, or a market in marine genetic resources—can reduce the second (also at a cost). Within a contract area both sources appear in both wedges, so neither instrument is confined to one margin there. Off contract, as above, only the second applies. I discuss these instruments further after the empirical analysis.

3.5 Mapping to the Empirical Design

The model maps naturally onto the regression design below. The contract-area indicator partitions observations into those where the permit is at stake ($M = 1$) and those where it is not ($M = 0$). Contractor fixed effects absorb firm-level variation in the returns and costs embedded in G and H, and in the compliance technology A, leaving the permit as the only thing that differs between a contractor's own tracts and the areas it surveys elsewhere. Under this mapping, the estimated coefficient $\hat{\beta}$ on the contract-area indicator recovers the precision gap $\tau_M - \tau_A$—the difference in species-reporting rates induced by the regulatory mechanism, holding contractor and year fixed. The counterfactual analysis below—which simulates non-contract-area reporting precision for every observation—corresponds to shutting down the $\phi > 0$ mechanism while holding all else fixed. The model also clarifies why sampling volume cannot explain the finding: Prediction 1 implies that prospectively-mined tracts have more samples, so any purely ecological or resource-constraints explanation would predict *better* species coverage there, not worse. The observed reversal is a signature of taxonomic deflation.

4 Methods

I test for taxonomic deflation by examining the taxonomic resolution of data reported on DeepData. Each observation is a biological specimen record submitted by a contractor to the ISA. When reporting to DeepData, contractors can populate up to eight taxonomic levels: Kingdom, Phylum, Class, Order, Family, Subfamily, Genus, and Species. In practice, most records stop short of the species, and the most common stopping point is the class. I focus on whether a record carries a species-level identification, since the species is the fundamental unit of biodiversity assessment (Claridge et al. 1997). Species-level identification is also a margin that the ISA's technical guidance leaves to contractor discretion. Its recommendations set no minimum taxonomic level, and they accept numbered morphospecies ("species 1, species 2") as "a good basis for baseline studies" (International Seabed Authority 2023). Consequently, a contractor that stops at a coarser level remains within the guidance. Taxonomic practice even supplies a standardized open-nomenclature vocabulary—signs such as "indet."—for recording identifications that stop short of a species (Horton et al. 2021), so coarse reporting reflects accepted scientific convention rather than obvious neglect. The dependent variable is therefore binary. It equals one if the record identifies the specimen at the species level, and zero otherwise. It should be noted that a species-level identification need not carry a formal name. Most deep-sea species have never been described, so taxonomists routinely assign specimens to provisional species, such as numbered morphospecies, and I count these identifications alongside named ones.[10] I compare species reporting in samples from prospectively-mined tracts to those from non-contract areas, accounting for the range of confounders the dataset allows.

The dataset contains over 160,000 biological specimen records from the CCZ, submitted by 12 mining contractors to DeepData. I classify each record as coming from a contract area or a non-contract area using the ISA's official shapefiles together with DeepData's own area label.[11] That is, a record inside the reporting contractor's own polygon is a contract-area record, and a record inside another contractor's polygon or an APEI is a non-contract-area record. This reclassifies 1,275 records that DeepData labels as outside but that

[10] Only 436 of the 5,578 species recorded in the CCZ have a formal name (Rabone, Wiethase, et al. 2023). The remainder are known by provisional designations, such as numbered morphospecies ("sp. 3") or molecular units ("Motu 4"), which separate one species from another without naming it. The ISA's recommendations accept this practice as a basis for baseline studies (International Seabed Authority 2023). Requiring a formal name would consequently measure how well described a group is, which a contractor does not control, and not how far the contractor took the identification. In practice, I code the outcome from the species field of DeepData, and I take each entry at face value unless it is plainly not an identification. That is, the outcome equals zero if the field is empty or if it holds a missing-value code (such as "indet."), a laboratory note (such as "no pcr product"), or the name of a higher group or a life stage (such as "Nematodes" or "larva"). It equals one for every other entry. I do not judge the quality of an identification, since doing so would require knowing what each contractor's laboratory means by a given entry. Of the 28,565 records coded as species-level, 74% carry a species name, 6% name two candidate species, 17% carry a numbered morphospecies, a molecular unit, or an open-nomenclature qualifier (such as "cf."), and 3% carry a placeholder (such as "sp.").

[11] DeepData labels each record as inside or outside the reporting contractor's exploration area. However, this label is not always reliable: Rabone, Horton, et al. (2023) note that some records labelled as outside in fact lie within the reporting contractor's claim. I instead let the shapefiles decide wherever a record's coordinates place it inside a mapped polygon.

fall inside their reporter's own polygon, and 8 records in the opposite direction. For the 801 records that fall outside every mapped polygon and the 10,508 records without coordinates, I rely on DeepData's label. The resulting comparison group contains 9,905 non-contract-area records, of which 92% lie inside another contractor's tract. Every record in the dataset, including those from non-contract areas, was submitted by a mining contractor; the dataset does not include samples from independent academic expeditions or other non-contractor sources. That fact is central to the empirical strategy—the comparison is between samples from different zone types reported by the same set of contractors, not between contractor data and independently-collected scientific data. I match each record to environmental data from Bio-ORACLE rasters (Tyberghein et al. 2012; Assis et al. 2018) using its reported longitude and latitude, yielding a matched dataset of 160,967 observations, of which 147,341 have complete environmental covariates and enter the specifications that add environmental controls. Table A1 in the Appendix reports summary statistics by zone type, including the number of records, year range, and species-reporting rates. Depth is taken from the contractor-reported `water_depth_m` field rather than from the bathymetry raster, and requires cleaning: the raw field mixes genuine depths with a sentinel code for missing values, a negative-down sign convention on a small number of records, and a handful of impossible magnitudes.[12]

I begin by showing the likelihood of species reporting in the raw data, then proceed to regression models that estimate the probability a specimen record includes a species identification. The main explanatory variable is whether the sample comes from a contractor-allocated exploration tract (a contract area). I present four specifications that incrementally add controls, showing how the estimated effect changes—and survives—as potential confounds are absorbed. I estimate a linear probability model (LPM) with heteroskedasticity robust standard errors: the object of interest is the average marginal effect of contract-area status on a binary outcome, and the linear model gives the most straightforward coefficient interpretation (Angrist and Pischke 2009). The first specification regresses the species-reporting indicator on an indicator for whether a sample comes from a contract area:

$$\text{Species Reported}_{ict} = \text{Intercept} + \beta \cdot \text{ContractArea}_i + \varepsilon_{ict} \tag{10}$$

The coefficient of interest, β, captures the difference in the mean probability of species-level identification between contract-area samples and non-contract-area samples, without accounting for confounders. Each

[12]Specifically, 2,673 records carry the value -9, a missing-data code; roughly ninety record depth as a negative number, which I recover as its absolute value after verifying that the magnitudes match the Bio-ORACLE bathymetry raster at the same coordinates to within about eighty meters; and a few dozen carry magnitudes of 21,600 and 44,000 meters, both deeper than any ocean. I retain values between 1,000 and 7,000 meters and set the rest to missing. The window is generous, as the CCZ abyssal plain runs roughly 3,500 to 5,500 meters and the shallowest genuine samples sit at about 1,850 meters on seamount flanks. After cleaning, reported depth correlates -0.94 with the independent bathymetry raster.

observation i is a biological specimen record from contractor c in year t, and ε_{ict} is the error term. A negative β indicates that specimens from contract areas are less likely to carry a species-level identification than specimens from non-contract areas.

The base specification omits potential confounders, so the estimated β could reflect factors other than strategic behavior. First, differences in taxonomic resolution could be driven by differences in scientific methodology rather than incentives. To absorb this, I add fixed effects for the mining contractor and for the year. Contractor fixed effects absorb average differences in reporting rates across the 12 contractors in the data—each a country-backed entity with an ISA exploration contract for one or more tracts (e.g., Nauru Ocean Resources Inc.). This matters because contractors that generally report more taxonomic detail might also be the ones submitting more non-contract-area samples, which would generate a spurious relationship in the raw data. Year fixed effects absorb average differences in reporting rates by year, accounting for improvements in scientific methods and taxonomic knowledge over time. The corresponding regression equations are:

$$\text{Species Reported}_{ict} = \beta \cdot \text{ContractArea}_i + \text{Contractor FE}_c + \varepsilon_{ict} \tag{11}$$

$$\text{Species Reported}_{ict} = \beta \cdot \text{ContractArea}_i + \text{Contractor FE}_c + \text{Year FE}_t + \varepsilon_{ict} \tag{12}$$

A second possible confounder is ecological. Non-contract areas and prospectively-mined tracts could differ in conditions that affect species-identification rates. The composition of the comparison group limits this concern, since 92% of non-contract-area records come from inside another contractor's tract, in the same nodule fields as the contract-area records. The remainder come from APEIs and from seabed outside any designated area. The ISA sited the APEIs to represent the environmental gradients of the CCZ (Wedding et al. 2013; Smith et al. 2021). However, the APEIs are not perfectly representative. Some of them differ from the contract areas they are meant to represent in nodule abundance, sediment characteristics, seafloor topography, and the flux of organic carbon to the seafloor (Washburn et al. 2021), and the APEI records in DeepData come from cooler and less productive waters than the contract-area records (Appendix Figure A5). To absorb these differences, I add four environmental controls that capture the major axes of variation across the CCZ: depth (which influences pressure, habitat type, and benthic community composition), sea surface temperature (the east-west and north-south productivity gradients), surface chlorophyll concentration (a direct measure of primary productivity and thus the organic carbon flux that sustains abyssal communities),

and surface dissolved oxygen (a further correlate of the same surface temperature and productivity regime). All environmental data are from Bio-ORACLE rasters matched to individual sample locations. Appendix Table A9 re-estimates this specification with near-bottom temperature, oxygen and current velocity substituted for the surface fields. Nodule abundance, another potentially relevant variable, is not included because spatially resolved nodule data are not publicly available at the sample level. However, the contractor fixed effects partially absorb nodule-related variation, since prospectively-mined tracts were allocated in large part on the basis of nodule density. The fourth regression equation is then:

$$\text{Species Reported}_{ict} = \beta \cdot \text{ContractArea}_i + \text{Environmental Controls}_i + \text{Contractor FE}_c + \text{Year FE}_t + \varepsilon_{ict} \quad (13)$$

where environmental controls include sea surface temperature, surface chlorophyll, surface dissolved oxygen, and depth matched to each sample's location. Three of the four are surface fields by design rather than by convenience. What structures abyssal benthic communities in the CCZ is the flux of particulate organic carbon reaching the seafloor, and that flux is set by surface productivity: species turnover across the eastern CCZ correlates with the flux of organic carbon to the seafloor (Bonifácio et al. 2021). Surface chlorophyll and sea surface temperature are therefore the reasonable proxies for the food supply that determines what lives on a given patch of seabed and how much there is to identify. Appendix Table A9 re-estimates the main specification with near-bottom temperature, oxygen and current velocity in place of the surface fields.

A remaining concern is that differences in species reporting could reflect differences in the *composition* of organisms sampled rather than strategic behavior. If contract areas happen to contain taxa that are inherently harder to identify to the species level, the gap could arise even without deflation. Importantly, taxonomy is itself endogenous to the exploration process: contractors who deflate will report coarser taxonomic classifications, so conditioning on taxonomy absorbs variation that may itself be generated by the deflation mechanism. Nonetheless, as a robustness exercise I present results from specifications that each condition on a single taxonomy fixed effect (phylum, class, order, family, or genus) on top of the full set of controls. Each specification restricts the sample to observations with a valid identification at the relevant taxonomic level. This is a conservative test: if the gap survives even after conditioning on endogenous taxonomy at the finest level (genus) it cannot be attributed to compositional differences in the organisms sampled at any taxonomic level, but it can also understate gaps in species reporting that are due to identifications that stopped at even higher taxonomic levels.

Two further robustness exercises exploit a structural feature of the dataset uncovered by overlaying official

ISA contract polygons on each observation's coordinates: a small but consistent share of biological samples are collected by one contractor inside *another* contractor's polygon, typically during joint expeditions or transit segments. These "guest" samples are processed by the same scientific institutions as own-tract samples, but carry no permit-related incentive for the sampler. The structure enables (i) a within-sampler design comparing each contractor's own-tract reporting to its guest-tract reporting (holding scientific capacity fixed) and (ii) a within-polygon design comparing owner-collected to guest-collected samples inside the same physical tract (holding ecology fixed). Together the two designs sandwich the headline effect from above and below using identifying variation that the contractor-and-environmental controls in the main regressions cannot use.

The four cross-zone specifications are reported as a ladder rather than collapsed into a single preferred model because each column trades off two desirable properties. The baseline column uses the full sample of contractor submissions and treats every observation symmetrically; in particular, it includes the substantial number of contractors who only ever submit data from inside their own contract tract, whose comparison group is mechanically other firms' guest visits to their tract and APEI material. These contractors contribute most of the variation in the dependent variable, but contribute none of the variation in the treatment indicator within their own observations. Successively adding contractor and year fixed effects and environmental controls forces the contract-area coefficient to be identified off increasingly demanding contrasts: within the same firm in the same year facing the same temperature, depth, and productivity. The cost is that the sample restricts and the within-firm comparison group shrinks. Reporting all four specifications lets the reader weigh that tradeoff directly; the headline effect is the within-firm-within-year-within-environment estimate (column 4), but the columns to its left document that the cross-firm comparison points in the same direction on the larger and more representative sample.[13]

Across all specifications, if contract-area samples are reported with lower taxonomic resolution after accounting for confounders, β should be negative and statistically significant.

5 Results

5.1 Exploration Intensity across Zones

The model's first prediction, that contractors conduct more physical sampling on prospectively-mined tracts than on non-contract areas, can be evaluated directly from the DeepData records before any regression

[13]The unit of observation throughout is the specimen record. Records from the same sampling station share their contract status and often their identification outcome, so Appendix Table A10 re-estimates the main specifications with each station as a single observation.

analysis. Figure 2 reports both the raw count of biological specimens by zone and the per-square-kilometer sampling intensity, using polygon areas computed from the ISA's official PMN and APEI shapefiles. Both views confirm Prediction 1: contract areas account for the overwhelming majority of specimens in absolute terms *and* receive sampling at a per-area rate that is hundreds of times higher than APEIs.

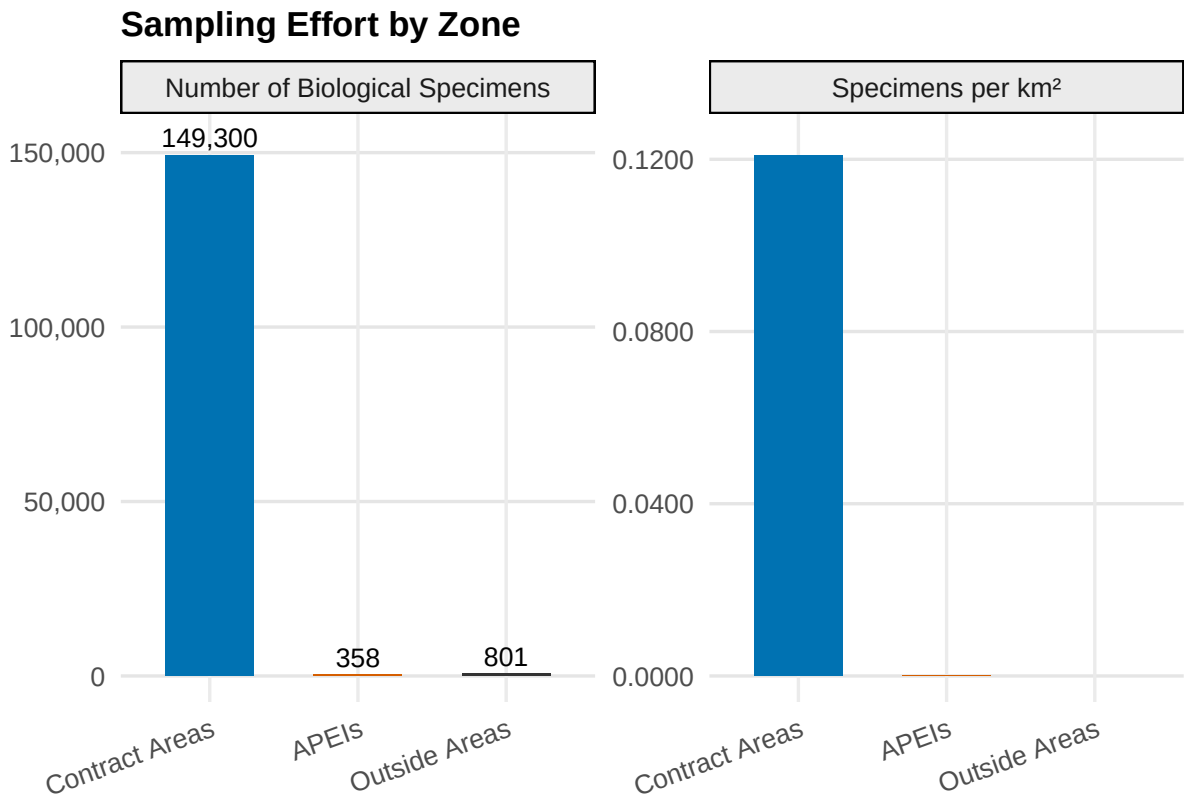


Figure 2: Sampling effort by zone, raw counts (left) and per-square-kilometer intensity (right). Sample counts are taken from the records in the analysis sample that report coordinates; areas are computed from the ISA's official PMN and APEI shapefiles projected to a cylindrical equal-area projection centered at 10°N. APEIs and contract areas have comparable aggregate footprint in the CCZ (∼2.0 vs. ∼1.2 million km^2), so the per-area panel is a like-for-like intensity comparison. The 'Outside' bar in the right panel is omitted because there is no well-defined denominator for samples falling outside both contract polygons and APEIs.

Consistent with Prediction 1, the overwhelming majority of biological samples in DeepData come from contract areas rather than non-contract areas, and the gap survives normalizing by the underlying footprint of each zone. Contract-area sampling intensity is 0.121 specimens per km^2 against 0.0002 per km^2 in APEIs—a ratio of roughly 663-to-one, even though contract areas and APEIs are of comparable aggregate

size in the CCZ (1,234,080 km^2 of PMN polygons against 1,962,328 km^2 of APEIs). This is what the model predicts: the cost of sampling is only worth paying for where it can result in a permit. Furthermore, this asymmetry shows up beyond the physical sampling. The administrative metadata fields that document the post-collection workflow (whether a taxonomist was involved, the identification method, status, and date) are filled in for roughly 70% of contract-area records but are only filled in for 1-6% of non-contract-area records, depending on the field. That means contractors invest substantially more effort in both field collection and compliance documentation on samples that factor into their own permit approval.[14] The critical question is whether this greater exploration effort translates into correspondingly greater taxonomic detail. Under a pure resource-constraints explanation, the zone with more samples should also have better species coverage. However, the model's second prediction is that the opposite holds.

5.2 Taxonomic Reporting by Zone

The raw data confirm Prediction 2. Figure 3 shows that most taxonomic levels are more likely to be reported in non-contract areas. For example, 47.5% of observations from non-contract areas report the species of the sample, against just 15.8% in contract areas. The raw difference could reflect many things—I show below that it persists after the obvious controls.

[14] Appendix Table A8 reports more details on the administrative metadata.

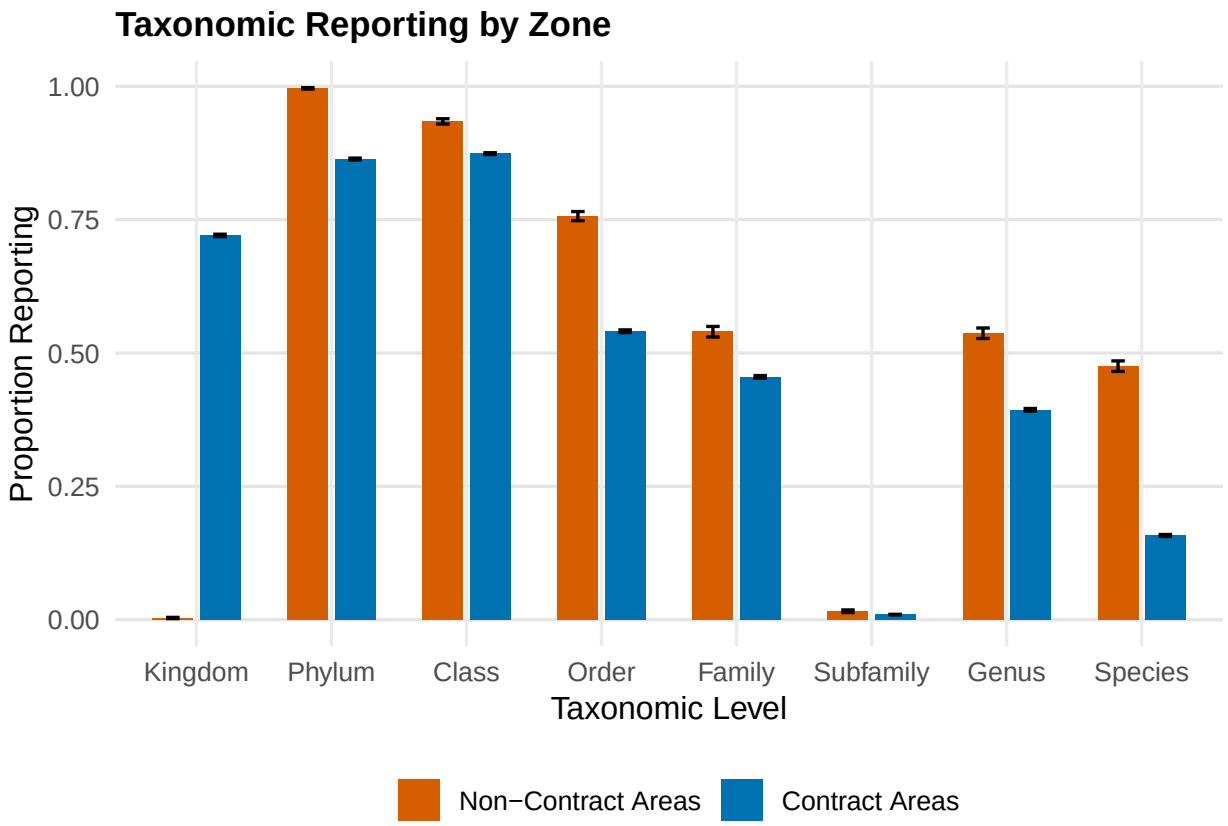


Figure 3: Proportion of specimen records with a non-missing value at each taxonomic level. Each level is evaluated independently from the others: a record counts as reported at a given level if and only if its corresponding taxonomic field (Kingdom, Phylum, ..., Species) contains a non-missing value, regardless of whether coarser or finer levels are also populated. The gap is largest at the species level, consistent with strategic deflation at the margin where species discovery poses the greatest regulatory risk. 95% confidence intervals shown.

5.3 Regression Results

The regressions provide a formal test of Prediction 2. Figure 4 plots the estimated $\hat{\beta}$—the precision gap $\tau_M - \tau_A$ from the model—and 95% confidence intervals for four specifications that incrementally add exogenous controls. The coefficient on contract-area status is negative and statistically significant in four of the four designs, implying anywhere from 32 to 11 percentage points lower probability of species reporting in contract areas relative to non-contract areas. The preferred specification (column 4, which adds environmental controls within firm and year) puts the gap at -0.113. Consistent with the model, contractor fixed effects—which absorb variation in scientific capacity across firms—reduce but do not eliminate the gap

(Appendix Figures A3 and A4 summarize sampling volume and species reporting by contractor). Adding year fixed effects shrinks the coefficient further ($p < 0.001$), but the gap remains negative and significant at conventional thresholds. Adding environmental controls reduces the coefficient to -0.113. However, this reduction comes from a change in the sample rather than from the controls themselves. Environmental data can only be matched to records with usable coordinates and depth, which excludes 13,626 records, and on that restricted sample the coefficient is already -0.092 before any controls are added. The controls themselves then move the estimate to -0.113, and do not change its sign. Most of the excluded records come from a single contractor, carry no coordinates, and contain no species-level identifications. Full regression results are reported in Table A2 in the Appendix. Because specimen records collected at the same sampling station may not be independent of each other, Appendix Table A10 also re-estimates these specifications treating each station as a single observation. Once contractor fixed effects are included, the station-level gap is of similar size to the record-level estimate. Appendix Tables A5 and A6 report a complementary exercise that replaces the binary species-reporting outcome with a graduated *stopping level* (0–7, the deepest taxonomic level each record reaches). Conditional on the same controls, contract-area records stop on average about 0.7 taxonomic levels shallower than non-contract-area records, implying the gap is not concentrated at the species threshold but rather it is a graduated tendency to stop earlier in the taxonomic hierarchy.

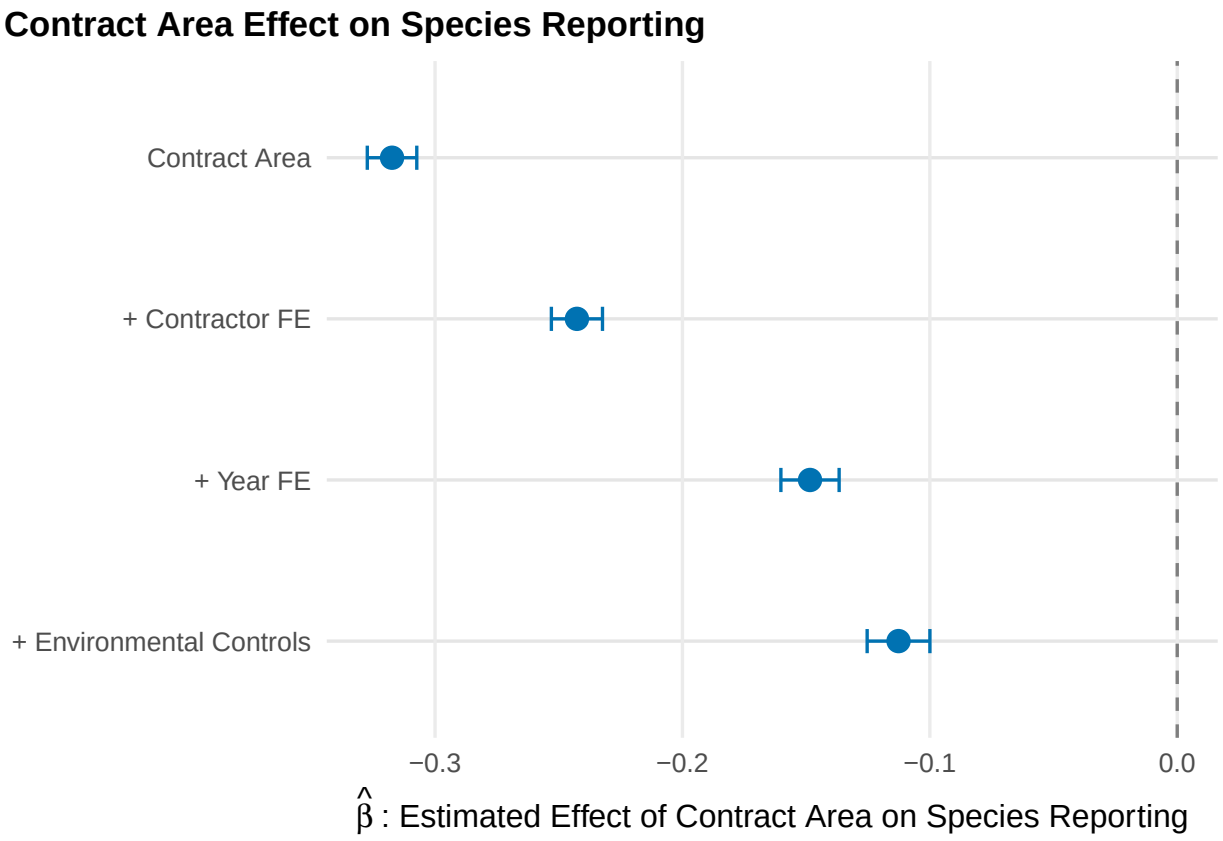


Figure 4: Estimated contract area coefficient ($\hat{\beta}$) from four LPM specifications, each adding incrementally more confounding variable controls. Each point represents the estimated difference in the probability of species-level identification for contract area samples relative to non-contract area samples. Negative coefficients indicate that species are less likely to be reported from a contract area than from a non-contract area. 95% confidence intervals computed with robust standard errors.

5.4 Robustness to Taxonomic Composition

A natural concern is that the gap reflects compositional differences in the organisms sampled across zones rather than strategic behavior. If contract areas happen to contain taxa that are inherently harder to identify, the reporting gap could arise without deflation. To address this, Figure 5 presents a taxonomy robustness exercise that separately conditions on each taxonomic level (from phylum through genus) on top of the full set of exogenous controls (contractor, year, and environmental variables). Taxonomy is itself endogenous to the exploration process—contractors who deflate report coarser taxonomic classifications—so conditioning on taxonomy is a conservative test that absorbs variation generated by the deflation mechanism itself. Even so, the contract-area coefficient remains negative and statistically significant when I compare organisms

within the same genus, ruling out compositional differences at every taxonomic level as an explanation for the gap (Table A3).

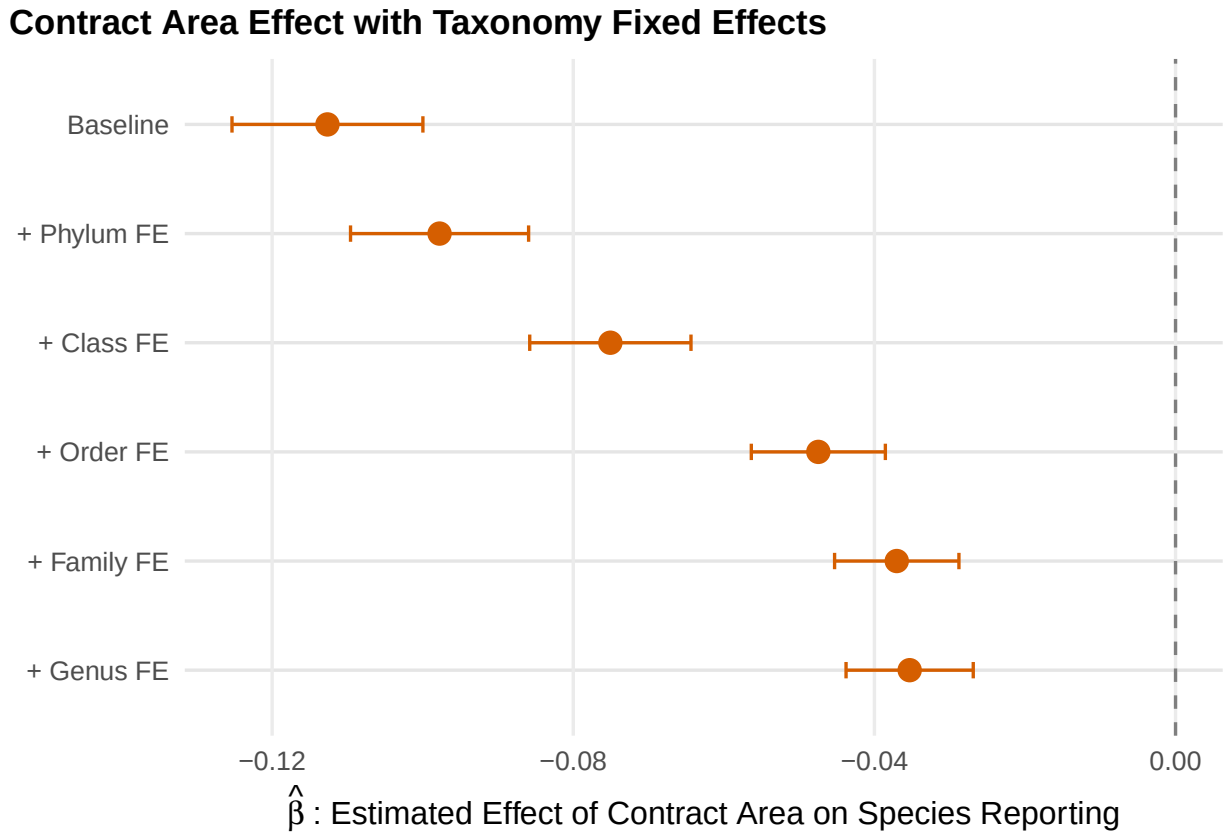


Figure 5: Estimated contract area coefficient ($\hat{\beta}$) from specifications that each add fixed effects for one taxonomic level. All specifications include contractor, year, and environmental controls. Each specification restricts the sample to observations with a valid identification at the relevant taxonomic level. 95% confidence intervals shown.

The persistence of $\hat{\beta} < 0$ after controlling for contractor identity, year, environmental conditions, and taxonomy indicates that the gap in species-level reporting between contract areas and non-contract areas is robust to a wide range of potential confounders. I take up the interpretation of this pattern, including alternative explanations, in the following sections. One important caveat is worth noting: species-level identification in the deep sea is genuinely difficult—an estimated 88–92% of CCZ metazoans are undescribed (Rabone, Wiethase, et al. 2023) and even expert teams identify fewer than 20% of observed species (Kennedy et al. 2019)—and I do not claim that every coarse identification reflects strategic behavior. But that difficulty is common to both zones, so it cannot explain the systematic *between-zone* gap. I take up that inherent

difficulty, and other non-strategic explanations, in the Discussion.

5.5 Within-Sampler and Within-Polygon Robustness

The contractor and taxonomy fixed effects above absorb cross-firm differences in scientific capacity and cross-taxon differences in identifiability, but they do not directly hold the *physical location* of each sample fixed. Geological and ecological conditions vary in space, and if contractor tracts are systematically located in regions where species are harder to identify for non-strategic reasons, the regression results above could be confounded. I address this by exploiting a structural feature of the dataset that emerges when I overlay the official ISA contract polygons onto each observation's coordinates: a small but consistent share of biological samples are collected by one contractor inside *another* contractor's polygon, typically during joint expeditions or transit segments. These "guest" samples are processed by the same scientific institution, in the same laboratory pipeline, and often during the same cruise as the sampler's own-tract samples, but they carry no permit-related incentive for the sampling contractor since the sampler does not hold mining rights at the location where the specimen was collected. Appendix Table A4 documents the full source decomposition.

This cross-contractor sampling enables two distinct robustness designs, each holding fixed a different confounder:

1. *Same contractor, different areas*: a within-sampler comparison that asks whether a given contractor reports samples differently when collecting in their own tract versus when collecting in another contractor's tract. Sampler fixed effects absorb cross-firm differences in scientific capacity, so the surviving variation isolates the regulatory regime change.

2. *Same area, different contractor*: a within-polygon comparison that asks whether, inside a single physical tract, samples collected by the polygon owner differ from samples collected by guest contractors. Polygon fixed effects absorb local ecology, so the surviving variation isolates the sampler-by-regime contrast at that location.

The two designs are complementary. The first holds reporting infrastructure fixed at the cost of leaving ecology to vary; the second holds ecology fixed at the cost of leaving reporting infrastructure to vary. Both deliver estimates that support the headline finding, and I report them in Tables 1 and 2 below. Both designs use as the sample all observations inside some PMN polygon, across all 12 sampling contractors. Samplers without Owner_i variation contribute nothing to the within-sampler $\hat{\beta}$, and polygons without Owner_i variation contribute nothing to the within-polygon $\hat{\beta}$, but those observations remain in the regression to inform the year and environmental coefficients. The estimating equations share a common structure:

$$\text{Species Reported}_{ipct} = \beta \cdot \text{Owner}_{ipct} + \mu_p + \gamma_c + \delta_t + X'_{ipct}\eta + \varepsilon_{ipct} \tag{14}$$

where $\text{Owner}_i = 1$ if observation i was collected by the polygon owner and 0 if by a guest. Different combinations of fixed effects identify β from different variation. With sampler fixed effects γ_c but no polygon fixed effects, β is the within-sampler difference between own-tract and guest-tract samples. With polygon fixed effects but no sampler fixed effects, it is the within-polygon difference between owner-collected and guest-collected samples. Including both is uninformative because each polygon has exactly one owner.

Table 1: Within-sampler design (same contractor, different areas): species reporting on Owner indicator, holding scientific capacity fixed via sampling-contractor fixed effects. Sample: all observations inside some PMN polygon, across all 12 sampling contractors. Samplers without Owner_i variation drop from $\hat{\beta}$ identification. Robust (HC1) standard errors in parentheses.

	Owner	+ Sampler FE	+ Year FE	+ Env. Controls
Intercept	0.467***	0.287***	1.475***	−32.036***
	(0.005)	(0.006)	(0.009)	(2.111)
Owner sampler	−0.297***	−0.236***	−0.106***	−0.092***
	(0.005)	(0.006)	(0.006)	(0.007)
Depth (m)				0.000***
				(0.000)
Sea Surface Temp.				0.551***
				(0.027)
Surface Chlorophyll				−7.513***
				(0.483)
Surface Dissolved Oxygen				3.820***
				(0.311)
Sampler FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Environmental Controls	No	No	No	Yes
Num.Obs.	149 300	149 300	149 300	146 603
R2	0.033	0.111	0.215	0.220

* p < 0.1, ** p < 0.05, *** p < 0.01

Within-sampler estimates are reported in Table 1. Column (1) reports the raw owner-vs-guest gap, $\hat{\beta} = -0.297$—here $\hat{\beta}$ is the coefficient on the Owner indicator, which plays the role of ContractArea in these designs, since an owner-collected sample is the permit-relevant one. Owner-collected specimens are roughly 30 percentage points less likely to carry a species-level identification than guest-collected specimens. Adding sampler fixed effects shrinks the estimate to −0.236, year fixed effects further to −0.106, and environmental controls leave it at −0.092 (SE 0.007). The within-sampler estimate is close to the main paper's preferred specification ($\hat{\beta} = -0.113$ in Figure 4 and column (4) of Table A2), and it is recovered from a more demanding source of variation, using only within-sampler, within-year variation between own-tract and guest-tract

Table 2: Within-polygon design (same area, different contractor): species reporting on Owner indicator, holding ecology fixed via polygon fixed effects. Sample: all observations inside some PMN polygon. Polygons without Owner_i variation drop from $\hat{\beta}$ identification. Robust (HC1) standard errors in parentheses.

	Owner	+ Polygon FE	+ Year FE	+ Env. Controls
Intercept	0.467***	0.283***	1.729***	−79.793***
	(0.005)	(0.008)	(0.011)	(1.918)
Owner sampler	−0.297***	−0.223***	−0.412***	−0.427***
	(0.005)	(0.008)	(0.007)	(0.007)
Depth (m)				0.000***
				(0.000)
Sea Surface Temp.				0.771***
				(0.025)
Surface Chlorophyll				−4.665***
				(0.502)
Surface Dissolved Oxygen				12.684***
				(0.326)
Polygon FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Environmental Controls	No	No	No	Yes
Num.Obs.	149 300	149 300	149 300	146 603
R2	0.033	0.093	0.214	0.221

* p < 0.1, ** p < 0.05, *** p < 0.01

samples.

Within-polygon estimates are reported in Table 2. Holding the physical tract fixed with polygon fixed effects alone yields $\hat{\beta} = -0.223$, roughly 2.4 times the within-sampler estimate; adding year fixed effects roughly doubles this further to −0.412, and environmental controls leave it at −0.427 (SE 0.007). The fully-controlled within-polygon estimate is roughly 5 times the within-sampler estimate. The expansion when I add year fixed effects reflects time variation in cross-sampler reporting practices—guest-tract samples are concentrated in particular years, and most of them are collected by a contractor with an above-average reporting rate, so within-polygon, within-year contrasts magnify the gap between owner and guest. The larger magnitude also reflects the fact that the within-polygon comparison picks up cross-sampler differences in scientific capacity at the same location alongside the regulatory regime change. Therefore, the fully-controlled within-polygon estimate is best read as an upper bound on the deflation effect under the assumption that residual within-polygon contractor-level heterogeneity reflects the regulatory regime rather than independent capacity differences. The two designs should be read as separate but mutually-reinforcing exercises. That is, the headline finding survives whichever confounder one is most worried about: holding scientific capacity fixed gives a −0.092 effect, holding ecology fixed gives a −0.427 effect. The within-sampler design plausibly understates the true deflation effect (guest-collected samples may receive less intensive post-processing than

own-tract samples regardless of regulatory regime); the within-polygon design plausibly overstates it (cross-firm differences in scientific capacity contribute alongside the regulatory mechanism).

The identifying variation behind the within-sampler estimate is visible in Figure 6, which plots, for each of the five contractors who appear as both polygon owners and as guests, the species-reporting rate of their own-tract samples against the species-reporting rate of their guest-tract samples. The pattern is consistent across firms with substantial guest-tract sample sizes: BGR reports 6% of its own-tract samples to species against 27% of its guest-tract samples (n=2,378); UKSRL reports 29% against 54% (n=6,553). The other three contractors with both roles (GSR, IFREMER, Korea) have very thin guest-tract samples (n=21 to 52) and should not be over-interpreted; for example, Korea's apparent reversal is identified off just 52 guest observations.

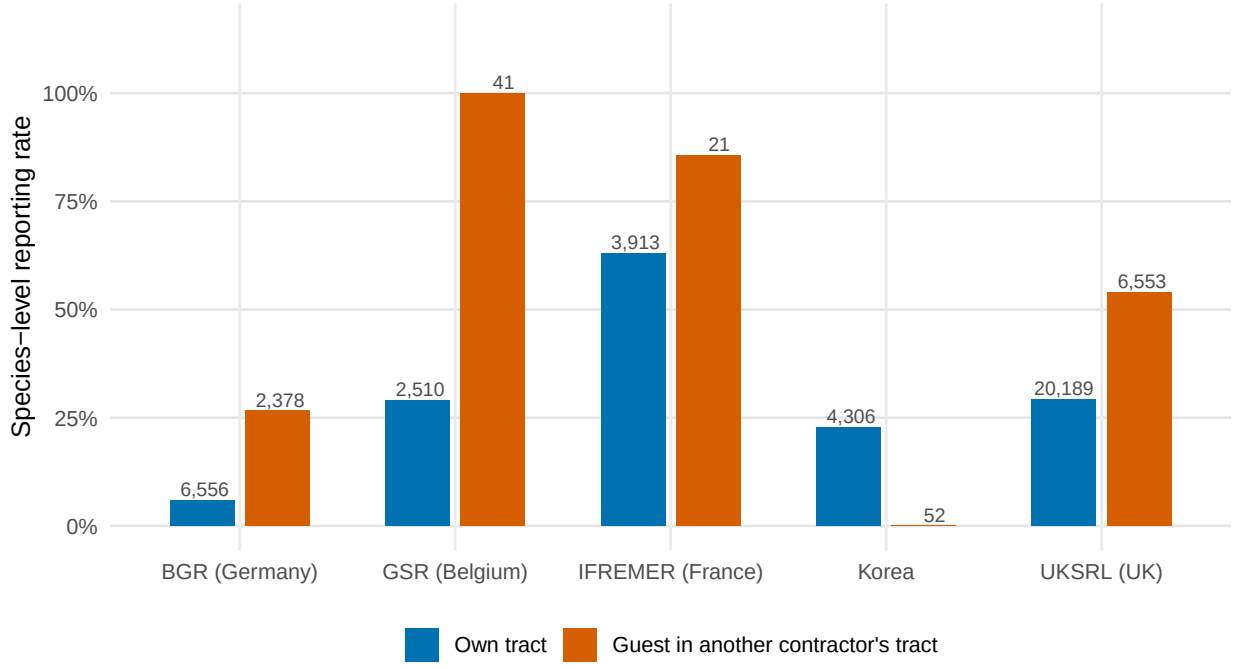


Figure 6: Species-level reporting rate, by sampling contractor and own-vs-guest status. Restricted to the five contractors who appear as both polygon owners and as guests in another contractor's polygon. Numbers above each bar are the underlying sample sizes; bars with very small n (the guest bars for GSR, IFREMER, and Korea) should be interpreted with caution. The within-sampler estimate in Table 1 averages these contrasts.

5.6 Mechanisms: Cruise-Level Selection, Sampling Gear, and the Identification Pipeline

The headline regression and the within-sampler and within-polygon designs document the existence of a deflation gap and rule out several confounders. They do not yet speak to the *channel* through which the gap operates: do contractors systematically organize their sampling and identification work differently for permit-relevant tracts? I examine three candidate channels that are available as data labels in DeepData: the expedition on which a specimen was collected, the gear that collected it, and the person who identified it. I then extend the third to the laboratory where the identification was made.

The first possible channel is the resources and methodology behind each given research cruise. Biological samples in DeepData are organized by expedition, each corresponding to a single research cruise with a particular vessel, scientific team, and operational window. The dataset spans 42 cruise legs in the analysis sample. If the deflation gap reflects between-expedition selection—contractors deploying different teams or methods on different cruises—then absorbing cruise-leg fixed effects should attenuate the gap.

The second is the sampling gear. What a specimen can be called depends on which animals the gear collects and what condition they arrive in. For example, a corer returns the small animals that live in the sediment, whereas a baited trap returns the scavengers that come to the bait. Many deep-sea "samples" are simply photographs from imaging transects with no specimen at all, which makes taxonomy very challenging (Horton et al. 2021). Gear choices are therefore consequential in imposing a ceiling on how precisely any record can ever be named: pooling across zones, species-level reporting rates by gear family run from 26% for multi/megacorer gear down to 4% for trap/baited. Gear is also one of the few fields regularly populated in all zones—a device is named for 99.8% of contract-area records and 99.1% of non-contract records—so the comparison does not rest on differential reporting of the control itself. I group devices into 7 families and test whether contractors systematically deploying different equipment on their own tracts can account for the measured taxonomic deflation.

The third and last is the identity of the taxonomist who produced each identification. 83% of records in the analysis sample report an identifier name, covering 83 distinct entries in the identifier field. Adding identifier fixed effects then asks whether the same identifier produces different reporting outcomes for contract-area and non-contract-area samples, necessarily holding individual expertise fixed by construction.

Table 3 reports the results.

Adding cruise-leg fixed effects to the main specification shrinks the contract-area coefficient from -0.113 (column 4 of Table A2) to -0.015 (SE 0.005). Roughly 86% of the headline effect is absorbed by between-

Table 3: Mechanism analyses. Baseline specification (Spec 4 of Table A2) adds cruise-leg fixed effects in column (1), sampling-gear-family fixed effects in column (2), and identifier-name fixed effects in columns (3)–(4). Column (4) restricts to identifiers who appear with both contract and non-contract observations. All specifications include contractor, year, and environmental controls. Robust (HC1) standard errors in parentheses.

	+ Cruise-leg FE	+ Gear-family FE	+ Identifier FE	+ Identifier FE (balanced)
Contract Area	−0.015***	−0.135***	0.024***	0.017***
	(0.005)	(0.007)	(0.005)	(0.005)
Contractor FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes
Cruise-leg FE	Yes	No	No	No
Gear-family FE	No	Yes	No	No
Identifier FE	No	No	Yes	Yes
Num.Obs.	147 341	147 341	133 750	19 395
R2	0.419	0.262	0.572	0.569

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

cruise variation: different expeditions sample different zones with different scientific teams, vessels, gear, and processing pipelines. Whether this between-cruise component is itself a confound to be controlled away or a *first stage* of the deflation mechanism is a matter of interpretation—contractors choose which cruises go where, with what teams and equipment, and those choices are themselves part of how the firm allocates identification effort across zones. The cruise-leg specification therefore does not test whether deflation is real, but it does bound the share of the headline gap that operates *conditional on* the contractor's cruise-level decisions. The within-cruise estimate of −0.015 is best read as a lower bound on the deflation that operates within a single expedition. That is, it estimates the part of the gap that cannot be attributed to cruise-level selection.

Gear choices are meaningful margins for taxonomic *precision*, but cannot explain taxonomic *deflation*. Among the firms that sample both inside and outside their own tracts, the gear mix on a firm's own tract differs from its gear mix elsewhere by a median dissimilarity index of 44% across the 3 firms with at least 100 records on each side (range 41–51). This means that roughly 44% of a typical firm's own-tract records would have to be re-collected with different equipment to reproduce the composition of its records from outside its tract. However, the empirical results show the gear choices are not responsible for taxonomic deflation—in fact, the coefficient measuring deflation grows from −0.113 to −0.135 (SE 0.007) once including fixed effects for gear choices (column 2 of Table 3). This is because the gear used in contract areas is, if anything, slightly better suited to species identification, so holding gear fixed removes a small composition effect that had been working against the gap.

To show this further, I calculate the species-reporting rate that each zone would have if gear alone determined

the reporting rate. I assign every record the average species-reporting rate of its gear family, pooled across zones, and then compare each zone's predicted rate against its actual rate. Based on gear alone, contract-area records in the analysis sample would be expected to report species 18.8% of the time, and they actually do so 16.7% of the time. Non-contract records would be expected to report species 15.7% of the time, which is slightly *lower* than in contract areas, but they actually do so 46.5% of the time. In other words, the two zones are sampled with gear of similar quality, but non-contract samples are identified to species at 3.0 times the rate their gear would predict, while contract-area samples fall slightly short of theirs.

Finally, I turn to the identifier-name fixed effects. On the sample of records that name an identifier, the contract-area coefficient is −0.061 without identifier fixed effects. With them included, it becomes positive: +0.024 (SE 0.005) on the full sample of records that name an identifier (column 3), and +0.017 (SE 0.005) when I restrict the sample to identifiers who appear in both zones (column 4). That is, on this sample the gap lies between identifiers rather than within them. Contract-area samples tend to be assigned to identifiers who report fewer species-level identifications, but a given identifier does not report contract-area samples less precisely. However, this should not be interpreted as evidence about how individual taxonomists behave, because this design cannot separate the taxonomist from the contractor that employs them. First, taxonomists rarely work for more than one contractor. Of the 83 named identifiers in the analysis sample, 71 work for exactly one contractor, and these account for 98% of the records that name an identifier. Identifier fixed effects therefore compare samples within the team that a contractor has assigned to its material, rather than across contractors. Second, whether an identifier is recorded at all depends on the contractor rather than on the individual record. Among contractors with at least five hundred own-tract records, the share of records that leave the identifier field empty is essentially zero for 3 contractors and 100% for 2 others, with the remaining contractors spread in between (Appendix Table A7).

The raw data show where the taxonomic precision gap sits, well before any fixed effects are added. Without controls, the gap is −0.30 across the analysis sample, −0.37 among records that name an identifier, and +0.02 among records that do not. That is, the gap is largest among the records where a taxonomist is named. This is the opposite of what one would expect if deflation were concentrated in records that no individual was willing to put their name to. Consequently, the identifier analysis supports a narrower conclusion than the one it would ideally test: the gap is associated with which identifiers handle a contractor's samples, and not with how a given identifier reports them. This is consistent with deflation operating through how a contractor organizes its identification work, rather than through the choices of individual taxonomists. However, it cannot quite be read as evidence that a given taxonomist reports identically across zones.[15]

[15] A within-identifier comparison would have to rely on the 20 of 83 identifiers who appear in both zones, who together account for 13% of the analysis sample. A single person accounts for 50% of those records, and two people account for 70%. In addition,

Since the identifier analysis cannot separate individuals from contractors, I next turn to the laboratories that perform the identifications. About 86% of records name an identification institution, and these cluster in a few labs with sharp specialties—foraminifera at one, nematodes at another, fishes and large crustaceans at a third, and so on. That specialization makes raw comparisons across institutions misleading, because taxa differ enormously in how often they can be named at all (nine in ten chordate records reach species level, against fewer than one in a hundred for corals or bryozoans). For example, a lab that works mostly on fish will post a high species rate whatever its effort because fish species are easier to identify for a variety of reasons. To isolate reporting behavior, the comparison has to hold the taxon fixed within the institution.

Table 4: The identifying institution and the deflation gap. Column (1) reproduces the baseline specification (Spec 4 of Table A2); columns (2)–(5) restrict to records that name an identification institution and progressively add institution and taxonomy fixed effects. Comparing specimens of the same taxon identified by the same institution (columns 4–5) isolates within-laboratory reporting. Robust (HC1) standard errors in parentheses.

	Baseline	Inst. named	+ Institution FE	+ Institution + Phylum FE	+ Institution + Genus FE
Contract Area	−0.113***	−0.002	0.038***	0.007	−0.006***
	(0.006)	(0.007)	(0.006)	(0.005)	(0.002)
Contractor FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes	Yes
Institution FE	No	No	Yes	Yes	Yes
Phylum FE	No	No	No	Yes	No
Genus FE	No	No	No	No	Yes
Num.Obs.	147 341	126 099	126 099	124 337	62 511
R2	0.217	0.308	0.547	0.574	0.853

* p < 0.1, ** p < 0.05, *** p < 0.01

The results are reported in Table 4. Restricting the sample to records that name an institution removes the gap: the contract-area coefficient is −0.002 on that sample, and +0.038 once institution fixed effects are added. As with identifiers, this partly reflects which contractors record the field, since the share of own-tract records that name an institution ranges from 0% to 100% across contractors. Holding the taxon fixed as well does not restore the gap. Comparing specimens of the same phylum identified by the same institution, the coefficient is +0.007, and comparing specimens of the same genus it is −0.006. That is, within the same laboratory and the same taxon, contract-area samples reach species level about as often as non-contract-area samples. I therefore find no evidence that a given laboratory identifies contract-area material less precisely. By contrast, among the records that do not name an institution, the coefficient is −0.069.

Taken together, these analyses narrow down the mechanisms behind taxonomic deflation, but do not isolate a single conclusive channel. Cruise-leg fixed effects absorb most of the gap, while sampling gear absorbs none of it. Identifier fixed effects absorb the gap among the records that name an identifier, and the gap

1 of the 20 is an institutional placeholder rather than a named individual. Unfortunately, this is too small a group to support a conclusion in either direction.

does not appear within laboratories. What these results have in common is the level at which they operate. Everything that absorbs part of the gap is a decision the *contractor* makes about how to organize a survey, such as which expedition goes where and which team identifies the samples. Within a given identifier or laboratory, contract-area samples are not identified less precisely. The most natural interpretation is therefore that the coarser data from contract areas result from how a contractor allocates its identification effort, rather than from individual scientists choosing to report less than they know. It should be noted, however, that the data cannot fully separate these two explanations, since taxonomists rarely work for more than one contractor and the recording of a taxonomist's name is itself a contractor-level practice. This distinction still matters for policy. If the gap arises from how contractors allocate identification effort, it can be addressed with instruments that take identification out of the contractor's hands (e.g., independent taxonomic review, sealed reporting, or certified-expert assignment). I discuss these options further in Section 7.

6 Discussion

The pattern I document fits the principal-agent problem described in the model: the ISA delegates biodiversity assessment to firms whose returns depend on subsequently being allowed to mine, but effort along the precision dimension is hard to monitor. As a result, contractors sample the tracts they intend to mine far more intensively than the non-contract areas, yet report those denser collections with systematically coarser taxonomy. More effort paired with less precision in the same place is the joint signature of this principal-agent problem, and it rules out any alternative explanation in which reporting quality is capped by resources or capacity. If that were the case, effort and precision would rise together rather than diverge. Two features of the empirical evidence are central to that strategic interpretation. First, the reporting gap survives the most demanding controls the data allow—contractor identity, year, environmental conditions, and even taxonomy down to genus. Second, both spatial robustness designs point in the same direction. When the *same* contractor samples in another firm's tract, it reports more precisely than in its own tract; when the *same* tract receives samples from both its owner and a guest contractor, the owner's samples are coarser. No non-strategic mechanism predicts both patterns. My empirical design documents an association rather than directly observing intent, but the alternatives I consider below cannot account for the joint within-sampler-and-within-polygon evidence.

6.1 Alternative Explanations

Two of the most obvious confounds are already dispatched by direct evidence above—compositional differences in the organisms sampled (the taxonomy-fixed-effects results, Figure 5) and spatial confounding (the within-sampler and within-polygon designs)—so I do not revisit them here. The alternatives that remain cannot be settled by a single specification, but I discuss below why I believe they are not responsible for the observed gap in taxonomic precision.

The most prominent objection is that the reporting gap reflects the inherent difficulty of deep-sea taxonomy rather than strategic behavior. Deep-sea identification is genuinely hard—less than 0.001% of the deep seafloor has been visually surveyed (Bell et al. 2025), expertise is scarce (Frutos et al. 2022), and an estimated 88–92% of CCZ species remain formally undescribed (Rabone, Wiethase, et al. 2023). Those constraints almost certainly explain why species-level reporting is low in *absolute* terms across the dataset.[16] What they cannot explain is why reporting rates differ *systematically* between zones, given that every sample in the analysis was collected from the abyssal CCZ and faces the same taxonomic knowledge gaps. If inherent difficulty were the sole driver, it would depress species reporting symmetrically across contract areas and non-contract areas. The gap between zones is precisely the component that science alone cannot explain.

A second alternative is that the gap reflects differences in *survey objectives* between zones. Non-contract-area surveys may be designed with biodiversity documentation as a primary goal and receive greater post-cruise taxonomic follow-up, while surveys of contract areas may prioritize environmental baseline characterization. If non-contract-area samples are more likely to be sent to specialized taxonomic laboratories or to receive extended analysis by academic collaborators, species-level identification rates could be higher even without any difference in field effort. Contractor fixed effects partially address this by absorbing average differences in each contractor's scientific methodology and institutional partnerships, but they cannot absorb *within-contractor* differences in how samples from different zones are processed after collection. If a contractor systematically invests more in the taxonomic analysis of its non-contract-area samples than its contract-area samples, that pattern would generate the observed gap. I cannot directly test for differential post-collection processing, but the direction of this explanation is itself consistent with the incentive story: a contractor that allocates more taxonomic effort to non-contract-area samples is responding to the same asymmetric

[16] Focused taxonomic work on CCZ material shows what species-level identification looks like when it is the purpose of the exercise. The Sustainable Seabed Knowledge Initiative workshop summarized in Jażdżewska and Horton (2026) described 24 new amphipod species (across ten families, one new family, and one new superfamily), and Bonifácio et al. (2021) identified deep-sea scale-worms to species using both morphology and DNA. Notably, much of this material was collected on contractors' own cruises, and Stewart et al. (2026) report a species-level dataset of sediment macrofauna from a study that a contractor funded on its own tract. Put simply, species-level identification of contract-area material is feasible when it is resourced. The 47.5% species-level reporting rate that I observe in the non-contract portion of DeepData is therefore in a reasonable range for this kind of work, and the much lower 15.8% rate in contract areas is well below what comparable effort would produce.

incentives the model describes.

A related possibility is that the gap reflects differences in *reporting timing* rather than deliberate underinvestment in taxonomy. Contractors may submit prospectively-mined tract data to DeepData promptly (before taxonomic analysis is complete) in order to satisfy compliance obligations, while non-contract-area samples carry no such pressure and receive more time for refined identifications before upload. If prospectively-mined tract records are submitted earlier in the taxonomic workflow, they would mechanically show lower species-level reporting even if the underlying scientific effort is identical. This is plausible, but it still reflects a difference in how contractors handle data from the two zones—the urgency to demonstrate compliance applies specifically to prospectively-mined tracts because of the permit at stake. That is, even under this interpretation, the asymmetry in reporting precision traces back to the asymmetry in incentives between zones. This is once again the same fundamental mechanism the model describes, just operating through the timing of data submission rather than the depth of taxonomic analysis. Furthermore, if compliance-driven early submission were the primary driver, one would expect contractors to update their records over time; the persistence of the gap suggests that such updates, if they occur, do not close it.

A further possibility is that the gap reflects differences in *sampling volume* rather than reporting incentives. Because contractors sample the tracts they intend to mine far more intensively than the non-contract areas (Appendix Figure A1), and because an operation that processes more material might identify each specimen more coarsely, the contract-area coefficient could in principle capture throughput rather than incentive. Adding controls for the number of records collected at each observation's location and on its research cruise leaves the contract-area gap in place; if anything the volume terms mostly cut against the concern, since records from more heavily sampled *locations* are reported *more* precisely, not less (only the cruise-level term points the other way, and it leaves the contract-area gap intact). A parallel control for the volume handled by each identifying institution can be computed only for records that name an institution, and on that subsample there is no gap to explain (Appendix Table A11). All told, sampling volume does not account for the zone gap.

Finally, the dependent variable measures taxonomic resolution as *reported in DeepData*, which may not perfectly reflect the underlying taxonomic work. Lower species-level reporting could reflect incomplete database uploads, delays in taxonomic revision, or administrative bottlenecks rather than reduced scientific effort. Rabone, Horton, et al. (2023) document significant data-quality issues in DeepData. These limitations are real, but they would have to affect contract-area and non-contract-area submissions *differentially* to explain the systematic gap I document. Symmetric data-quality issues add noise but cannot generate a systematic bias in one direction. Furthermore, if for some reason species from contract areas were reported

at higher taxonomic resolutions outside of DeepData, this could still suggest some strategic behavior on the part of mining contractors, as they would be deflating the species reporting to the ISA relative to data published to the wider scientific community.

6.2 Counterfactual: Reporting without Taxonomic Deflation

The model's second prediction is that contractors could identify organisms more precisely without sampling more, but rationally choose not to—a deflation gap that the welfare analysis then prices as a precision wedge. To quantify the gap, I run a counterfactual exercise that shuts down the $\phi > 0$ mechanism while holding exploration effort and all other covariates fixed. Figure 7 plots the predicted number of specimens that would include a species entry in the absence of taxonomic deflation, out of the 147,341 specimens with complete environmental covariates (the sample on which all four specifications can generate predictions). I generate fitted values from the regressions with the contract-area indicator set to zero for every specimen. That is, I simulate the probability of species reporting as if every specimen were reported with the same taxonomic precision as non-contract-area samples. I use the largest and smallest estimated $|\hat{\beta}|$ to bound the result. As Figure 7 shows, removing the deflation would lead to substantially higher predicted reporting—between 42,779 and 70,033 specimens would report the species of the sampled organism, against just 27,449 in actuality.

The counterfactual also decomposes the paper's motivating question. Of the 119,892 specimens in the sample that lack a species identification, the share attributable to strategic incentives ranges from 13% (15,330 specimens) under the most conservative coefficient to 36% (42,584 specimens) under the largest. The complementary shares—87% and 64% respectively—would still lack a species identification even absent any deflation, representing the genuine limits of deep-sea taxonomy: undescribed fauna, specimen damage, and scarce expertise that no policy can immediately overcome. Even by the most conservative estimate, strategic incentives account for more than ten thousand missing species identifications in DeepData; under the largest estimate, more than forty thousand. This exercise is in partial-equilibrium—it holds exploration intensity and all covariates fixed, varying only the reporting precision. The discovery hazard deters sampling as well as precision, as the exploration wedge in equation (9) shows, so removing it would raise exploration too; the counterfactual, which holds exploration fixed, should therefore be interpreted as a lower bound on the full informational welfare gain from removing the incentive distortion. The headline coefficient is the relevant input for this exercise even though the mechanism section locates much of the gap in between-cruise and between-identifier variation. Those specifications *attribute* the deflation to particular channels (which cruises a contractor sends where, which identifier processes which sample) but they do not eliminate it. The

contractor's choice of cruise composition and identifier assignment is itself part of how the firm responds to the permit at stake, so the contractor's total behavioral response (the relevant object for the social planner) is captured by the headline estimate, not by the within-channel residual.

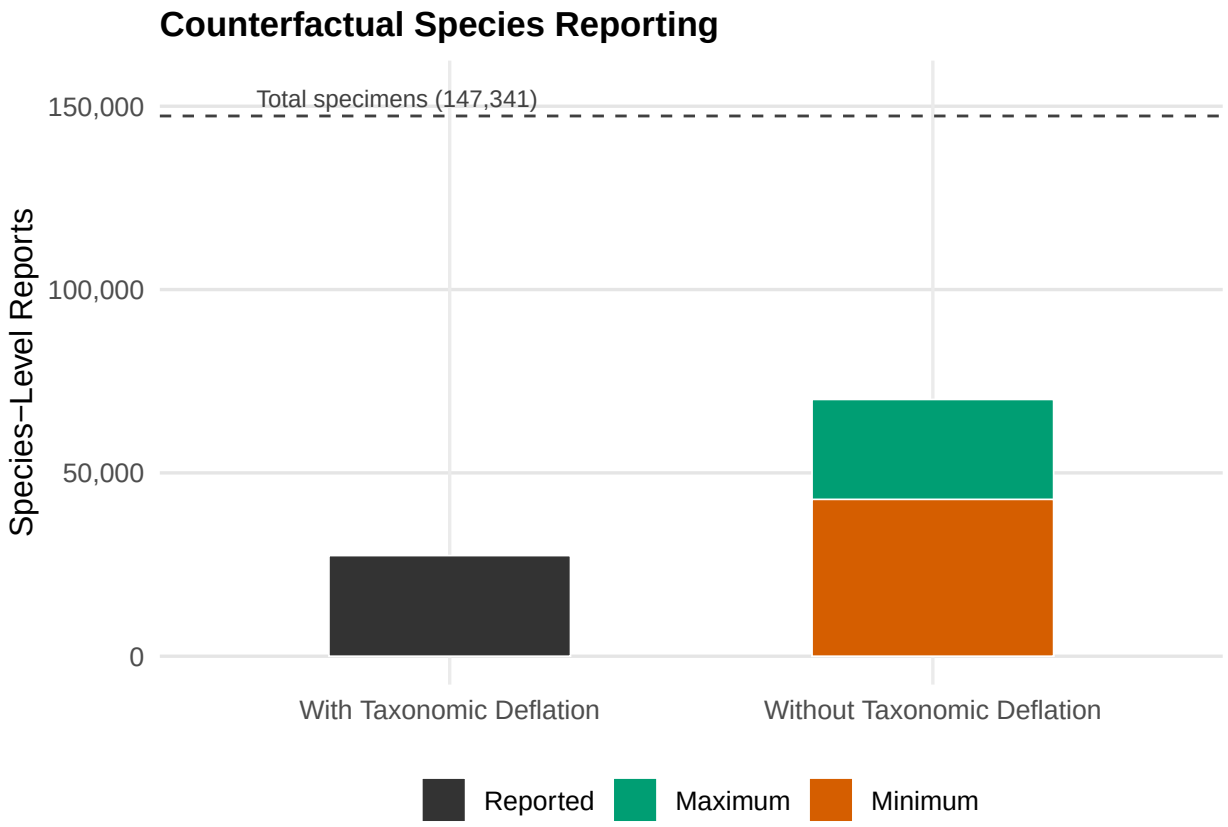


Figure 7: The dark bar plots observed species reports. The vermillion bar plots the counterfactual using the most conservative estimated contract area effect; the green bar extends above it to show the counterfactual using the largest estimated effect. The dashed reference line marks the total number of specimens in the sample: the gap between the counterfactual bars and this line is the share of missing species identifications that taxonomic deflation cannot explain.

6.3 Counterfactual: Exploration Coverage in the APEIs

The precision wedge is only half of the social cost the model identifies. The second is an exploration wedge: because no permit rides on ground the contractor will never mine, its sampling there is pinned by the scientific return alone, and APEIs are barely sampled at all. Figure 2 documented the raw asymmetry: prospectively-mined tracts are sampled roughly 663 times as densely as APEIs, even though the two zones cover comparable area in the CCZ. The second counterfactual therefore asks what would have come from

the missing exploration. The density of prospectively-mined tracts is a natural reference because it is the exploration the ISA's permit incentive actually elicits. However, it should be noted that according to the model, even the exploration on prospectively-mined tracts is underprovided, so this should be thought of as a lower bound. Nevertheless, scaling APEI sampling to the level of prospectively mined tracts is a move toward closing the exploration wedge, even if not all the way. I take the sampling density of prospectively-mined tracts (0.121 specimens per km^2) and apply it to the 1,962,328 km^2 of APEIs, holding the APEIs' own species-reporting rate (34.4%) fixed so that the exercise varies only sampling volume and not reporting precision. Under this scenario (Figure 8), the 358 specimens actually recorded inside APEIs become roughly 237,000 (about 87,000 physical sampling events), and the 123 species-level identifications they currently carry become on the order of 82,000—an additional 81,000 species-level identifications from the areas specifically set aside to safeguard biodiversity.

This exercise should be viewed as a thought experiment, not a forecast. The extrapolation is enormous: it scales 358 actual specimens up by a factor of roughly 663, and the 34.4% reporting rate that anchors the species-report projection is itself estimated from only those 358 specimens. The sampling-noise band alone spans 70,000–93,000 reports. Furthermore, prospectively mined tract density is not a normative benchmark: APEIs are set aside precisely so they will not be mined, and there is no reason the socially optimal survey intensity for a reference area should exactly equal that of a commercial tract. Still, this thought experiment conveys an order of magnitude. The exploration that the contractor regime leaves undone outside the reach of a permit is not a rounding error against the biodiversity record: in fact, the unsampled coverage of the APEIs alone dwarfs the entire identified APEI collection.

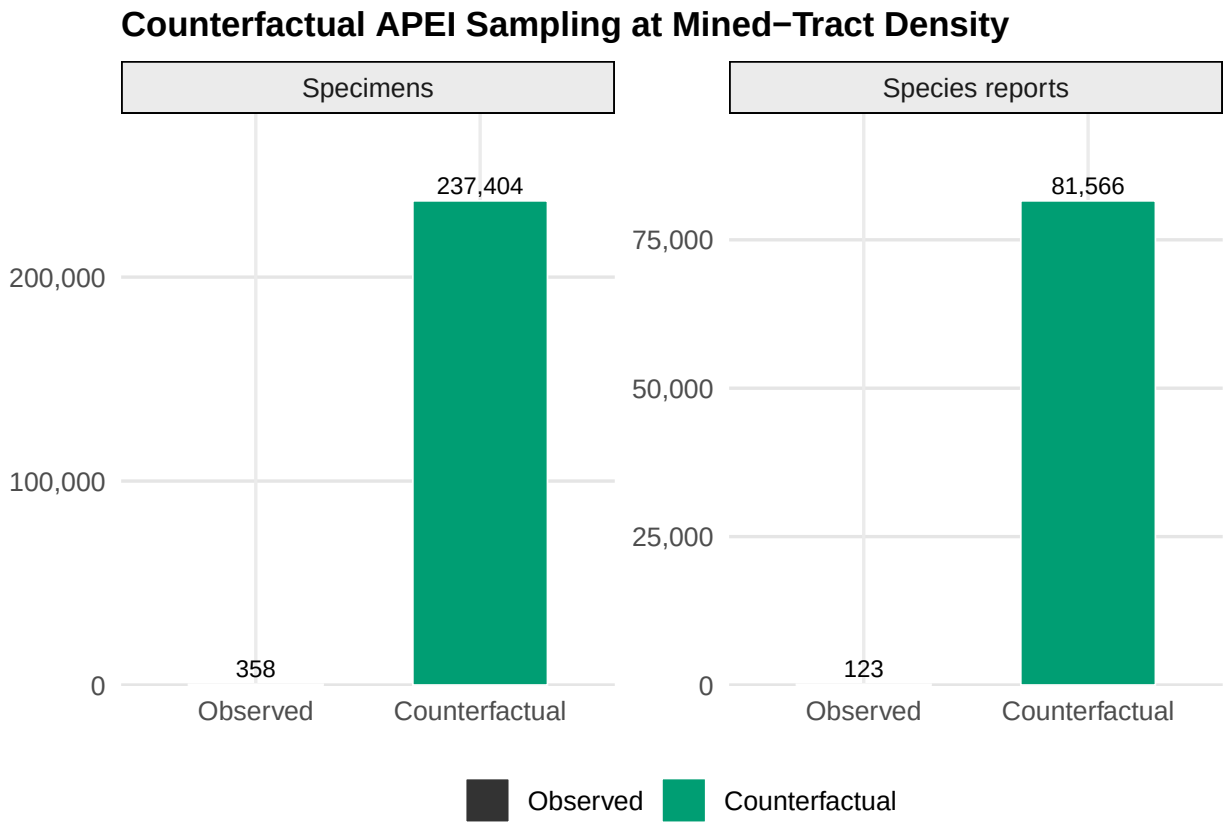


Figure 8: Observed versus counterfactual APEI sampling. The dark bars plot what APEIs actually yield: the specimens recorded inside them and the species-level identifications those specimens carry. The green bars plot the counterfactual in which APEIs are sampled at the per-area density of prospectively-mined tracts, holding the APEIs' own species-reporting rate fixed so that only sampling volume changes. The counterfactual is a deliberately extreme thought experiment—a several-hundredfold extrapolation from a small observed base—and is meant to bound the order of magnitude of the exploration wedge, not to predict any particular survey outcome.

7 Policy Options

The findings above do not suggest that the private requirement to report data is hopelessly flawed. The current arrangement may still be preferable to either a scenario where the seabed is mined without exploration or a scenario where the seabed is explored extensively by a third party prior to mining. Under the first, society loses scientific knowledge, genetic resources, and the opportunity to protect particular discoveries. Under the second, the third party tasked with exploration does not internalize the returns to speed and may

indefinitely delay mining in service of perfect exploration. An ideal system balances the incentive to explore and the incentive to mine against the social optimum. Since the social benefits of exploration are not well-compensated, the ISA's solution of using the opportunity to mine as a carrot to incentivize exploration may well be the best that can be done, in spite of the perverse incentives I document here. This is true even under the more skeptical view that deep-sea mining may not be economically justified once full environmental costs are included (Sumaila et al. 2023): if mining ultimately does not proceed, the biodiversity data generated under the exploration requirement may prove to be the regime's most lasting contribution. What the analysis does show is that the regime has a meaningful principal-agent problem. The mining company internalizes the opportunity cost of delaying mining, but it does not act as a perfect agent of the ISA on the exploration margin, because it does not place equivalent weight on scientific discovery.

If the problem is significant enough to warrant policy changes, the ISA has several options. The exploitation regulations remain under negotiation, with numerous outstanding issues unresolved (Pickens et al. 2024), so there is an opportunity to address deflation incentives before the regulatory framework is finalized. The model's welfare analysis identifies two distinct wedges, the precision wedge and the exploration wedge, and each has its own set of possible policy remedies.

The two wedges likely differ in their costs to close. Shrinking the precision wedge need not require any additional exploration. Many of the specimens whose identifications are recovered in the counterfactual of Section 6 have already been collected, sorted, and stored, so all that is missing is expert attention to material already in hand. The cost is therefore taxonomists' time rather than ship time. Of course, some changes in sampling methodology may also be required to close the gap fully, in which case not all of the work could be done retrospectively. However, the finding that gear type is not primarily responsible for the gap suggests that much of it could be. Closing the exploration wedge is a much larger undertaking. Sampling the APEIs at the density of prospectively-mined tracts would mean collecting roughly 237,000 specimens, which is more than the entire biological record for the CCZ, assembled over roughly two decades. A rough accounting of the survey effort already in DeepData puts the vessel time behind it in the tens of millions of dollars, and the counterfactual would require a similar amount again.[17]

The data also indicate which of the two inputs is the binding constraint. Every contractor pays roughly the same rate for ship time. However, the number of specimen records needed to document one distinct species ranges from 3.6 for COMRA (China) to 661 for NORI (Nauru), a difference of roughly 182 to one. This difference largely reflects how much identification work is done after the samples are collected. In effect,

[17] The survey effort behind the database (575 sampling days across 52 cruise legs) implies vessel time of roughly $20–90 million at research-vessel day rates. The range depends on how transit time is counted and on how much of each cruise is attributed to biological sampling rather than resource assessment.

contractors are already paying for the more expensive input (ship time) while limiting the cheaper one (identification). This is consistent with the mechanism results in Section 5.6, and it implies that the most cost-effective interventions are those that fund taxonomic work directly, rather than those that require more sampling. Among the remedies discussed below, this favors those that change *who* does the identification over those that tighten what contractors are required to report.

The intuitively desirable remedy for the precision wedge is to identify the margins on which contractors can shirk and strengthen the reporting requirement against them. The structure of the Mining Code gives some notion of where to begin. The obligation to identify organisms precisely currently appears only in the Legal and Technical Commission's non-binding recommendations (International Seabed Authority 2023). Moving a minimum-resolution requirement into the binding rules could therefore turn a discretionary margin into an enforceable one. The draft exploitation framework already separates legally binding "Standards" from advisory "Guidelines" (International Seabed Authority 2019), which makes it a natural vehicle for such a rule. However, this is easier said than done, because many different margins can admit deflation. For example, a rule specifying which sampling gear contractors must use would likely not help, since the gap remains conditional on sampling gear (Section 5.6). Because the results imply that non-contract areas are reported with greater taxonomic precision than contract areas, more attention to how exploration standards differ between zone types could be fruitful. The ISA should also take care to prevent the gaming of any updated rules. A requirement that explicitly compares samples from contract and non-contract areas could create an incentive to underreport from non-contract areas as well, while a requirement that explicitly targets specific taxonomic levels may induce false specificity. This is a version of Goodhart's law: once a measure becomes a target, it ceases to be a good measure.

A more targeted approach to the precision wedge would be to decouple discovery from denial. As the model shows, the contractor reduces taxonomic precision below the socially optimal level because precise identification risks triggering conservation review. Several concrete mechanisms could reduce the regulatory cost of a species discovery. *Safe harbor provisions* would guarantee that a contractor's mining approval cannot be revoked solely on the basis of species discoveries made during its own survey, which removes the direct penalty for precise identification. A similar option, *sealed reporting*, would require contractors to submit their raw biological specimens and taxonomic data to an independent scientific body, which would conduct or verify the taxonomic analysis separately from the permitting process. The ISA would receive the final taxonomic assessments only after the permitting decision has been made, or would receive them in an aggregated or anonymized form that cannot be traced to specific permit applications. This breaks the link between what the contractor discovers and what the regulator can use against it. Contractors would

have no reason to suppress precise identifications, because those identifications could not influence their own permit outcome. The independent body would still make the full taxonomic record publicly available, which preserves the scientific value of the data even if conservation becomes less targeted. These mechanisms address the incentive distortion by reducing the regulatory penalty of a species discovery. However, they could undermine conservation goals by limiting the ISA's ability to adapt management to ecological discoveries, which may be untenable.

A further remedy is to change *who* does the science. Rather than relying on the contractor's incentives, the ISA (or a third party it funds) could carry out the biological exploration and identification directly. An alternative explorer has no permit at stake, so this approach removes the deflation incentive at its source. In principle, it also addresses both wedges at once: independent surveys would supply the exploration that contractors underprovide, and independent identification would supply the precision that they suppress. The taxonomic capacity to identify CCZ fauna at high resolution exists, albeit sparsely. What is missing is the mandate and the funding to apply it. The ISA has begun a limited version of this approach through its Sustainable Seabed Knowledge Initiative (SSKI), which funds the description of new deep-sea species (International Seabed Authority 2022; Jażdżewska and Horton 2026). However, because SSKI works on material that has already been collected, it currently addresses only the precision wedge. The same is true of *independent taxonomic review*, which would assign post-collection identification to third-party specialists who are outside the influence of mining contractors. This would improve precision, but it would still leave sample collection, and its timing and methodology, to the contractor.

Public provision would also help with enforcement. Independent baseline surveys of non-contract areas would fill the exploration gap, particularly in the APEIs, which remain largely unsurveyed (Section 5). They would also provide an external benchmark against which contractor submissions could be judged. That is, a regulator with its own representative sample of CCZ fauna could tell when a contractor's reported species rate falls well short of what comparable areas yield. In this way, an independent reference collection would serve as a tool for detecting taxonomic deflation, and not only as a supplement to the biodiversity record. However, such a program would require careful design, so that it penalizes genuine shirking without punishing firms for the inherent difficulty and randomness of deep-sea biological sampling.

To address the exploration wedge, the ISA could reward discoveries. It could do so directly, by awarding prizes or otherwise allowing contractors to monetize discoveries, or indirectly, by strengthening the link between high-effort exploration and mining approval. The instruments that affect both margins at once are those that pay for a *named specimen*, such as a prize per described species, an advance commitment, or access to genetic-resource markets. This is because the return they create depends on sampling and precision

together, rather than on either one alone. In the model, this return corresponds to the social value s. Making the contractor internalize it raises both sampling and precision in non-contract areas. The outcome would then be closer to the economic optimum, in which a single agent internalizes all of the relevant costs and benefits of both mining and exploration. By contrast, an instrument that raised the return to resolution alone would change precision without changing sampling. However, these instruments would be difficult to design well, and they could backfire if they end up rewarding contractors for continuing their current practice.

It should also be noted that, on a tract the contractor holds, the two types of instrument (those that reduce the regulatory penalty and those that reward discovery) are not confined to separate margins. There, both the regulatory penalty and the scientific return enter both wedges, so either type of instrument affects exploration and precision alike. Outside the contract areas the division is clearer. No permit depends on what is found there, so the regulatory penalty plays no role—all that remains is the unpriced scientific value. This is why the shortfall in APEI sampling can only be addressed by instruments that reward discovery.

These policy directions connect to the model's further implications. First, if deflation intensifies with tract value, monitoring and enforcement resources should be concentrated on the highest-value exploration contracts. Second, the model cautions against assuming that tighter precautionary standards will reduce deflation on their own. Raising the regulatory cost of a discovery lowers precision directly, but it also discourages the sampling that would put a discovery at risk, making the net effect ambiguous. This is one reason to pair tightening with mechanisms that decouple discovery from denial.

Several of the remedies above are not specific to ocean resources or to areas beyond national jurisdiction. They have counterparts in other regulatory settings where the regulated firm produces the information that the regulator uses to oversee it. Independent taxonomic review has analogues in blind grading, third-party financial audits, and the assignment of clinical-trial endpoints to independent adjudicators. An external benchmark follows the same logic as monitoring by environmental NGOs, independent inspector general offices, and academic researchers. Rewards for discovery correspond to data prizes, publication incentives, and other instruments that compensate the provision of information. The closest parallel is endangered-species regulation. The ESA creates similar perverse incentives for private landowners who might harbor listed species (Brown and Shogren 1998; Langpap et al. 2018), and it gives them little incentive to cooperate with species-information collection (Polasky and Doremus 1998). In that context, safe-harbor agreements have been shown to mitigate preemptive habitat destruction (Byl 2019), and the design of takings compensation substantially shapes landowners' incentives to conserve rather than develop habitat (Innes et al. 1998).

However, the deep-sea mining setting differs in one important respect. Landowners under the ESA already hold property rights over their land, while mining contractors do not own the seabed and are applying to use a resource considered the common heritage of humankind. There is also less legal architecture for granting property rights over biological discoveries in the deep sea, which limits the applicability of the compensation-based solutions that have proven effective on land. This makes the design of reporting incentives all the more important. In addition, the entry into force of the Agreement on Biodiversity Beyond National Jurisdiction (BBNJ) introduces new obligations around environmental impact assessment and marine genetic-resource benefit-sharing, and these interact with the ISA's existing regime in ways that remain unresolved (Willaert and Soete 2025). If the BBNJ framework creates additional regulatory consequences for species discoveries, it could strengthen rather than weaken the incentive to deflate.

Each of these options carries costs and risks. All of them involve shifting or adding costs onto the ISA. In addition, any program that funds new exploration directly brings back the problem that the current arrangement was designed to avoid, namely an explorer who does not internalize the opportunity cost of delaying mining. Changing the institutional arrangement therefore involves real tradeoffs. Without action, however, taxonomic deflation threatens the scientific integrity of deep-sea biodiversity assessments.

8 Conclusion

The imprecision of deep-sea mining data is not fully explained by scientific difficulty. That difficulty is real, but it is symmetric across areas that could be explored. This paper identifies an asymmetry in species reporting that follows permit incentives, which is the signature of taxonomic deflation. The counterfactual decomposition implies that even by the most conservative estimate, strategic incentives account for roughly 13% of the missing species identifications in DeepData—around 15,000 specimens under that bound, and as many as 43,000 under the largest—all of which would carry a species-level identification if discovery did not threaten the permit. Furthermore, because no permit is at stake in non-contract areas, they go almost entirely unsurveyed. APEIs are sampled roughly 663 times less densely than prospectively-mined tracts, and had they been explored at mined-tract density they would have yielded on the order of 81,000 additional species-level records rather than the 123 they now contain. The underlying reality is that no single entity has fully aligned incentives to balance exploration and exploitation of the deep sea, and the ISA's operative solution—using the prospect of mining as a carrot for biodiversity reporting—inevitably creates the precision and exploration wedges I document. Whether those wedges are small enough to tolerate or large enough to fix is a choice the exploitation regulations now under negotiation will have to grapple with. Those choices will determine whether the deep-sea biodiversity record reflects what the data could show or only what

mining contractors find it convenient to report.

References

Angrist, Joshua David, and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics : An Empiricist's Companion*. Princeton University Press.

Ardron, Jeff A. 2018. "Transparency in the Operations of the International Seabed Authority: An Initial Assessment." *Marine Policy* 95 (September): 324–31. https://doi.org/10.1016/j.marpol.2016.06.027.

Arrow, Kenneth J., and Anthony C. Fisher. 1974. "Environmental Preservation, Uncertainty, and Irreversibility." *The Quarterly Journal of Economics* 88 (2): 312–19. https://doi.org/10.2307/1883074.

Assis, Jorge, Lennert Tyberghein, Samuel Bosch, Heroen Verbruggen, Ester A. Serrão, and Olivier De Clerck. 2018. "Bio-ORACLE V2.0: Extending Marine Data Layers for Bioclimatic Modelling." *Global Ecology and Biogeography* 27 (3): 277–84. https://doi.org/10.1111/geb.12693.

Bell, Katherine L. C., Kristen N. Johannes, Brian R. C. Kennedy, and Susan E. Poulton. 2025. "How Little We've Seen: A Visual Coverage Estimate of the Deep Seafloor." *Science Advances* 11 (19): eadp8602. https://doi.org/10.1126/sciadv.adp8602.

Bennear, Lori S., and Sheila M. Olmstead. 2008. "The Impacts of the 'Right to Know': Information Disclosure and the Violation of Drinking Water Standards." *Journal of Environmental Economics and Management* 56 (2): 117–30. https://doi.org/10.1016/j.jeem.2008.03.002.

Blanchard, Catherine, Ellycia Harrould-Kolieb, Emily Jones, and Michelle L. Taylor. 2023. "The Current Status of Deep-Sea Mining Governance at the International Seabed Authority." *Marine Policy* 147 (January): 105396. https://doi.org/10.1016/j.marpol.2022.105396.

Bonifácio, Paulo, Lenka Neal, and Lénaïck Menot. 2021. "Diversity of Deep-Sea Scale-Worms (Annelida, Polynoidae) in the Clarion-Clipperton Fracture Zone." *Frontiers in Marine Science* 8. https://doi.org/10.3389/fmars.2021.656899.

Britz, Ralf, Anna Hundsdörfer, and Uwe Fritz. 2020. "Funding, Training, Permits—the Three Big Challenges of Taxonomy." *Megataxa* 1 (1): 49–52. https://doi.org/10.11646/megataxa.1.1.10.

Brown, Gardner M., and Jason F. Shogren. 1998. "Economics of the Endangered Species Act." *Journal of Economic Perspectives* 12 (3): 3–20. https://doi.org/10.1257/jep.12.3.3.

Byl, Jacob P. 2019. "Perverse Incentives and Safe Harbors in the Endangered Species Act: Evidence From Timber Harvests Near Woodpeckers." *Ecological Economics* 157 (March): 100–108. https://doi.org/10.1016/j.ecolecon.2018.11.008.

Claridge, Michael F., Hassan A. Dawah, and Michael R. Wilson, eds. 1997. *Species : The Units of Biodiversity*. The Systematics Association Special Volume Series. Chapman & Hall.

de Marchi, Scott, and James T. Hamilton. 2006. "Assessing the Accuracy of Self-Reported Data: An Evaluation of the Toxics Release Inventory." *Journal of Risk and Uncertainty* 32 (1): 57–76. https://doi.org/10.1007/s10797-006-6666-3.

Delić, Teo, Peter Trontelj, Michal Rendoš, and Cene Fišer. 2017. "The Importance of Naming Cryptic Species and the Conservation of Endemic Subterranean Amphipods." *Scientific Reports* 7 (1): 3391. https://doi.org/10.1038/s41598-017-02938-z.

Doshi, Anil R., Glen WS Dowell, and Michael W. Toffel. 2013. "How Firms Respond to Mandatory Information Disclosure." *Strategic Management Journal* 34 (10): 1209–31.

Duflo, Esther, Michael Greenstone, Rohini Pande, and Nicholas Ryan. 2013. "Truth-Telling by Third-Party Auditors and the Response of Polluting Firms: Experimental Evidence from India." *The Quarterly Journal of Economics* 128 (4): 1499–545.

Frutos, Inmaculada, Stefanie Kaiser, Łukasz Pułaski, Maciej Studzian, and Magdalena Błażewicz. 2022. "Challenges and Advances in the Taxonomy of Deep-Sea Peracarida: From Traditional to Modern Methods." *Frontiers in Marine Science* 9. https://doi.org/10.3389/fmars.2022.799191.

George, Anna, and Richard Mayden. 2005. "Species Concepts and the Endangered Species Act: How a

Valid Biological Definition of Species Enhances the Legal Protection of Biodiversity." *Natural Resources Journal* 45 (March): 369–407.

Ghezelayagh, Ava, Jeffrey W. Simmons, Julia E. Wood, et al. 2025. "Comparative Species Delimitation of a Biological Conservation Icon." *Current Biology* 35 (2): 398–406.e4. https://doi.org/10.1016/j.cub.2024.11.053.

Glover, Adrian G, Helena Wiklund, Chong Chen, and Thomas G Dahlgren. 2018. "Managing a Sustainable Deep-Sea 'Blue Economy' Requires Knowledge of What Actually Lives There." *eLife* 7 (November): e41319. https://doi.org/10.7554/eLife.41319.

Gonzalez-Lira, Andres, and Ahmed Mushfiq Mobarak. 2021. "Slippery Fish: Enforcing Regulation When Agents Learn and Adapt." *National Bureau of Economic Research Working Paper Series* No. 28610. https://doi.org/10.3386/w28610.

Harrington, Winston. 1988. "Enforcement Leverage When Penalties Are Restricted." *Journal of Public Economics* 37 (1): 29–53. https://doi.org/10.1016/0047-2727(88)90003-5.

Henry, Claude. 1974. "Investment Decisions Under Uncertainty: The "Irreversibility Effect"." *The American Economic Review* 64 (6): 1006–12.

Heyes, Anthony, and Neil Rickman. 1999. "Regulatory Dealing – Revisiting the Harrington Paradox." *Journal of Public Economics* 72 (3): 361–78. https://doi.org/10.1016/S0047-2727(98)00098-X.

Horton, Tammy, Leigh Marsh, Brian J. Bett, et al. 2021. "Recommendations for the Standardisation of Open Taxonomic Nomenclature for Image-Based Identifications." *Frontiers in Marine Science* 8. https://doi.org/10.3389/fmars.2021.620702.

Innes, Robert, Stephen Polasky, and John Tschirhart. 1998. "Takings, Compensation and Endangered Species Protection on Private Lands." *Journal of Economic Perspectives* 12 (3): 35–52. https://doi.org/10.1257/jep.12.3.35.

International Seabed Authority. 2013. *Decision of the Council Relating to Amendments to the Regulations on*

Prospecting and Exploration for Polymetallic Nodules in the Area and Related Matters. ISBA/19/C/17. International Seabed Authority.

International Seabed Authority. 2015. *Recommendations for the Guidance of Contractors on the Content, Format and Structure of Annual Reports*. ISBA/21/LTC/15. International Seabed Authority, Legal and Technical Commission.

International Seabed Authority. 2019. *Draft Regulations on Exploitation of Mineral Resources in the Area*. ISBA/25/C/WP.1. International Seabed Authority, Legal and Technical Commission.

International Seabed Authority. 2020. *Report of the Workshop on Deep-Sea Taxonomic Standardization: Strategic Approaches for Collaboration*. International Seabed Authority.

International Seabed Authority. 2022. *Sustainable Seabed Knowledge Initiative*.

International Seabed Authority. 2023. *Recommendations for the Guidance of Contractors for the Assessment of the Possible Environmental Impacts Arising from Exploration for Marine Minerals in the Area*. ISBA/25/LTC/6/Rev.3. International Seabed Authority, Legal and Technical Commission.

Isaac, Nick J. B., James Mallet, and Georgina M. Mace. 2004. "Taxonomic Inflation: Its Influence on Macroecology and Conservation." *Trends in Ecology & Evolution* 19 (9): 464–69. https://doi.org/10.1016/j.tree.2004.06.004.

Jaeckel, Aline L. 2017. *The International Seabed Authority and the Precautionary Principle: Balancing Deep Seabed Mineral Mining and Marine Environmental Protection*. Brill | Nijhoff. https://doi.org/10.1163/9789004332287.

Jażdżewska, Anna M., Karolina Biniek, Pedro Martínez Arbizu, and Annemiek Vink. 2025. *Hidden Behind the Scene: High Diversity, Low Connectivity of Deep-Sea Amphipoda in the Polymetallic Nodule Fields in the Northeast Pacific*. EGUsphere [preprint]. https://doi.org/10.5194/egusphere-2025-1794.

Jażdżewska, Anna M., and Tammy Horton. 2026. "New Deep-Sea Amphipoda from the Clarion-Clipperton Zone: 24 New Species Described Under the Sustainable Seabed Knowledge Initiative: One Thousand

Reasons Campaign." *ZooKeys* 1274 (March): 1–16. https://doi.org/10.3897/zookeys.1274.176711.

Jones, Daniel O. B., Maria Belen Arias, Loïc Van Audenhaege, et al. 2025. "Long-Term Impact and Biological Recovery in a Deep-Sea Mining Track." *Nature* 642 (8066): 112–18. https://doi.org/10.1038/s41586-025-08921-3.

Jones, Daniel O. B., Jennifer M. Durden, Kevin Murphy, et al. 2019. "Existing Environmental Management Approaches Relevant to Deep-Sea Mining." *Marine Policy* 103 (May): 172–81. https://doi.org/10.1016/j.marpol.2019.01.006.

Kennedy, Brian R. C., Kasey Cantwell, Mashkoor Malik, et al. 2019. "The Unknown and the Unexplored: Insights Into the Pacific Deep-Sea Following NOAA CAPSTONE Expeditions." *Frontiers in Marine Science* 6 (August). https://doi.org/10.3389/fmars.2019.00480.

Kennedy, Brian R. C., and Randi D. Rotjan. 2023. "Mind the Gap: Comparing Exploration Effort with Global Biodiversity Patterns and Climate Projections to Determine Ocean Areas with Greatest Exploration Needs." *Frontiers in Marine Science* 10 (November). https://doi.org/10.3389/fmars.2023.1219799.

Kim, Eun-Hee, and Thomas P. Lyon. 2011. "Strategic Environmental Disclosure: Evidence from the DOE's Voluntary Greenhouse Gas Registry." *Journal of Environmental Economics and Management* 61 (3): 311–26. https://doi.org/10.1016/j.jeem.2010.11.001.

Krusberg, Tilde, Lova Schildt, Jean-Baptiste Jouffray, Erik Zhivkoplias, and Robert Blasiak. 2024. "A Review of Marine Genetic Resource Valuations." *Npj Ocean Sustainability* 3 (1): 46. https://doi.org/10.1038/s44183-024-00081-7.

Kürzel, Karlotta, Stefanie Kaiser, Anne-Nina Lörz, et al. 2022. "Correct Species Identification and Its Implications for Conservation Using Haploniscidae (Crustacea, Isopoda) in Icelandic Waters as a Proxy." *Frontiers in Marine Science* 8. https://doi.org/10.3389/fmars.2021.795196.

Langpap, Christian, Joe Kerkvliet, and Jason F. Shogren. 2018. "The Economics of the U.S. Endangered Species Act: A Review of Recent Developments." *Review of Environmental Economics and Policy* 12 (1): 69–91. https://doi.org/10.1093/reep/rex026.

Levin, Lisa A., Diva J. Amon, and Hannah Lily. 2020. "Challenges to the Sustainability of Deep-Seabed Mining." *Nature Sustainability* 3 (10): 784–94. https://doi.org/10.1038/s41893-020-0558-x.

Lueck, Dean, and Jeffrey A. Michael. 2003. "Preemptive Habitat Destruction Under the Endangered Species Act." *The Journal of Law and Economics* 46 (1): 27–60. https://doi.org/10.1086/344670.

Lugthart, Peter. 2026. "The Environmental Costs and Consumer Benefits of Competitive Smog Check Markets." *Job Market Paper*, October.

Lyon, Thomas P., and John W. Maxwell. 2011. "Greenwash: Corporate Environmental Disclosure Under Threat of Audit." *Journal of Economics & Management Strategy* 20 (1): 3–41. https://doi.org/10.1111/j.1530-9134.2010.00282.x.

Milgrom, Paul, and Chris Shannon. 1994. "Monotone Comparative Statics." *Econometrica* 62 (1): 157–80. https://doi.org/10.2307/2951479.

Mu, Yingfei, Edward Rubin, and Eric Yongchen Zou. 2026. "What's Missing in Environmental Self-Monitoring: Evidence from Strategic Shutdowns of Pollution Monitors." *The Review of Economics and Statistics* 108 (3): 597–612. https://doi.org/10.1162/rest_a_01477.

Niner, Holly J., Jeff A. Ardron, Elva G. Escobar, et al. 2018. "Deep-Sea Mining With No Net Loss of Biodiversity—An Impossible Aim." *Frontiers in Marine Science* 5. https://doi.org/10.3389/fmars.2018.00053.

Oliva, Paulina. 2015. "Environmental Regulations and Corruption: Automobile Emissions in Mexico City." *Journal of Political Economy* 123 (3): 686–724. https://doi.org/10.1086/680936.

Pickens, Chris, Hannah Lily, Ellycia Harrould-Kolieb, Catherine Blanchard, and Anindita Chakraborty. 2024. "From What-If to What-Now: Status of the Deep-Sea Mining Regulations and Underlying Drivers for Outstanding Issues." *Marine Policy* 169 (November): 105967. https://doi.org/10.1016/j.marpol.2023.105967.

Polasky, Stephen, and Holly Doremus. 1998. "When the Truth Hurts: Endangered Species Policy on Private

Land with Imperfect Information." *Journal of Environmental Economics and Management* 35 (1): 22–47. https://doi.org/10.1006/jeem.1998.1021.

Rabone, Muriel, T Horton, D O B Jones, E Simon-Lledó, and A G Glover. 2023. "A Review of the International Seabed Authority Database DeepData from a Biological Perspective: Challenges and Opportunities in the UN Ocean Decade." *Database* 2023 (January): baad013. https://doi.org/10.1093/database/baad013.

Rabone, Muriel, Joris H. Wiethase, Erik Simon-Lledó, et al. 2023. "How Many Metazoan Species Live in the World's Largest Mineral Exploration Region?" *Current Biology* 33 (12): 2383–2396.e5. https://doi.org/10.1016/j.cub.2023.04.052.

Smith, Craig R., Malcolm R. Clark, Erica Goetze, Adrian G. Glover, and Kerry L. Howell. 2021. "Editorial: Biodiversity, Connectivity and Ecosystem Function Across the Clarion-Clipperton Zone: A Regional Synthesis for an Area Targeted for Nodule Mining." *Frontiers in Marine Science* 8. https://doi.org/10.3389/fmars.2021.797516.

Stewart, Eva C. D., Helena Wiklund, Lenka Neal, et al. 2026. "Impacts of an Industrial Deep-Sea Mining Trial on Macrofaunal Biodiversity." *Nature Ecology & Evolution* 10 (2): 318–29. https://doi.org/10.1038/s41559-025-02911-4.

Sumaila, U. R., L. Alam, K. Pradhoshini, et al. 2023. "To Engage in Deep-Sea Mining or Not to Engage: What Do Full Net Cost Analyses Tell Us?" *Npj Ocean Sustainability* 2 (1): 19. https://doi.org/10.1038/s44183-023-00030-w.

Topkis, Donald M. 1978. "Minimizing a Submodular Function on a Lattice." *Operations Research* 26 (2): 305–21. https://www.jstor.org/stable/169636.

Turner, Erick H., Annette M. Matthews, Eftihia Linardatos, Robert A. Tell, and Robert Rosenthal. 2008. "Selective Publication of Antidepressant Trials and Its Influence on Apparent Efficacy." *New England Journal of Medicine* 358 (3): 252–60. https://doi.org/10.1056/NEJMsa065779.

Tyberghein, Lennert, Heroen Verbruggen, Klaas Pauly, Charles Troupin, Frederic Mineur, and Olivier De

Clerck. 2012. "Bio-ORACLE: A Global Environmental Dataset for Marine Species Distribution Modelling." *Global Ecology and Biogeography* 21 (2): 272–81. https://doi.org/10.1111/j.1466-8238.2011.00656.x.

Washburn, Travis W., Daniel O. B. Jones, Chih-Lin Wei, and Craig R. Smith. 2021. "Environmental Heterogeneity Throughout the Clarion-Clipperton Zone and the Potential Representativity of the APEI Network." *Frontiers in Marine Science* 8.

Wedding, L. M., A. M. Friedlander, J. N. Kittinger, et al. 2013. "From Principles to Practice: A Spatial Approach to Systematic Conservation Planning in the Deep Sea." *Proceedings of the Royal Society B: Biological Sciences* 280 (1773): 20131684. https://doi.org/10.1098/rspb.2013.1684.

Willaert, Klaas, and Anemoon Soete. 2025. "The Interaction Between the BBNJ Agreement and the International Deep Sea Mining Regime: More Questions Than Answers?" *Ocean Development & International Law* 56 (2): 176–213. https://doi.org/10.1080/00908320.2025.2472634.

Zakota, Mark. 2025. "EPA Scrutiny and Voluntary Environmental Disclosures." *Review of Accounting Studies* 30 (4): 3514–57. https://doi.org/10.1007/s11142-025-09901-8.

Zhivkoplias, Erik, Jean-Baptiste Jouffray, Paul Dunshirn, Agnes Pranindita, and Robert Blasiak. 2024. "Growing Prominence of Deep-Sea Life in Marine Bioprospecting." *Nature Sustainability* 7 (8): 1027–37. https://doi.org/10.1038/s41893-024-01392-w.

Appendix

The appendix is organized into three sections. **Extra Tables** and **Extra Figures** present material that is referenced in the main text but not shown there. **Additional Analyses** presents further exercises that address specific concerns about the main results.

Extra Tables

Table A1: Summary statistics for biological specimen records in the DeepData database. Contract Areas are records from the reporting contractor's own tract; APEIs are records located inside an Area of Particular Environmental Interest; Outside Areas are all other records, most of which fall inside another contractor's tract. Species reporting rate is the proportion of records that include a species-level identification.

Zone	Specimen Records	Contractors	Year Range	Species Reporting Rate
Contract Areas	151,062	12	2004–2021	15.8%
APEIs	358	6	2015–2018	34.4%
Outside Areas	9,547	6	2010–2021	48%
Total	160,967	12	2004–2021	17.7%

Table A2: Linear probability model estimates of species-level reporting. Dependent variable: indicator for species-level taxonomic identification. Comparison group is non-contract areas (records from outside the reporting contractor's own tract). Robust standard errors in parentheses.

	Contract Area	+ Contractor FE	+ Year FE	+ Environmental Controls
Intercept	0.475***	0.554***	1.219***	−10.659***
	(0.005)	(0.025)	(0.036)	(1.842)
Contract Area	−0.317***	−0.243***	−0.148***	−0.113***
	(0.005)	(0.005)	(0.006)	(0.006)
Depth (m)				0.000***
				(0.000)
Sea Surface Temp.				0.254***
				(0.020)
Surface Chlorophyll				−9.198***
				(0.468)
Surface Dissolved Oxygen				0.918***
				(0.288)
Contractor FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Environmental Controls	No	No	No	Yes
Num.Obs.	160 967	160 967	160 967	147 341
R2	0.040	0.128	0.221	0.217

* p < 0.1, ** p < 0.05, *** p < 0.01

Table A3: Taxonomy fixed effects robustness. Dependent variable: species-level identification. Comparison group is non-contract areas (records from outside the reporting contractor's own tract). All specifications include contractor, year, and environmental controls. Each specification restricts the sample to observations with a valid identification at the relevant taxonomic level. Robust standard errors in parentheses.

	Baseline	+ Phylum FE	+ Class FE	+ Order FE	+ Family FE	+ Genus FE
Intercept	−10.659***	−3.099*	−3.371**	0.050	−3.055	−2.648*
	(1.842)	(1.771)	(1.689)	(1.965)	(1.953)	(1.471)
Contract Area	−0.113***	−0.098***	−0.075***	−0.047***	−0.037***	−0.035***
	(0.006)	(0.006)	(0.005)	(0.005)	(0.004)	(0.004)
Contractor FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes	Yes	Yes
Phylum FE	No	Yes	No	No	No	No
Class FE	No	No	Yes	No	No	No
Order FE	No	No	No	Yes	No	No
Family FE	No	No	No	No	Yes	No
Genus FE	No	No	No	No	No	Yes
Num.Obs.	147 341	136 601	129 298	83 889	71 854	63 316
R2	0.217	0.320	0.563	0.666	0.767	0.820

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Sample Source Composition

Each biological specimen record in DeepData is reported by a specific mining contractor and is associated with a specific physical location. I use the official ISA contract polygon shapefiles, downloaded from the ISA's Maps repository at `isa.org.jm/exploration-contracts/maps/`, to match each observation to the contract polygon (if any) that contains it. Combining this with the identity of the *sampling* contractor (extracted from the `area_key` field) yields four mutually exclusive observation types:

- **Own tract**: the observation falls inside the sampling contractor's own PMN polygon. This is the setting in which a permit is at stake.
- **Other tract**: the observation falls inside *another* contractor's PMN polygon. For example, a German (BGR) sample may be collected within Singapore's (OMS) tract during a joint or transit cruise. The sampling contractor has no permit-related incentive at this location.
- **APEI**: the observation falls inside an Area of Particular Environmental Interest, where mining is prohibited.
- **Outside all**: the observation falls in international waters outside any contract polygon and outside any APEI.

This decomposition is useful for two reasons. First, it describes where the biological samples in the dataset

come from. Second, the four types differ in their regulatory exposure. Table A4 reports counts and species-level reporting rates for each type. Roughly 93% of CCZ specimens are collected in the sampling contractor's own tract, 6% are collected by one contractor inside another contractor's tract, and observations in APEIs and outside all polygons together make up less than 1%. Species-level reporting is much lower for own-tract samples than for the other three types. Specimens collected by a contractor within its own tract are identified to species only 17.0% of the time. However, when contractors sample inside *another* contractor's tract, they report at the species level 46.7% of the time. In APEIs, where mining is prohibited and no commercial interest is at stake, 34.4% of specimens are identified to species. The gap between own-tract and other-tract samples, which come from the same physical region, motivates the within-sampler and within-polygon exercises reported in Tables 1 and 2 of the main text.

Table A4: Composition of biological observations by spatial source. "Own tract" = inside the sampling contractor's own PMN polygon; "Other tract" = inside another contractor's PMN polygon; "APEI" = inside an Area of Particular Environmental Interest; "Outside all" = outside any PMN polygon or APEI. Spatial classification uses official ISA shapefiles.

Sample source	Observations	Share of total	Species-level reporting
Own tract	140,255	93.2%	17.0%
Other tract	9,045	6.0%	46.7%
APEI	358	0.2%	34.4%
Outside all	801	0.5%	41.6%

Stopping-Level Analysis

The main outcome in the paper is binary: whether a record includes a species-level identification. This outcome does not distinguish between records that stop at different levels above species. Some records are identified to genus, while others stop at family, order, or class. In this subsection, I recode the outcome to capture the level at which each identification stops.

For each record, I construct a *stopping level* that takes integer values from 0 to 7: 0 if no taxonomic field is populated, 1 if the record stops at kingdom, 2 at phylum, 3 at class, 4 at order, 5 at family, 6 at genus, and 7 if the record reaches species. Table A5 reports the distribution of stopping levels by zone, and Table A6 reports a linear regression of the stopping level on the contract-area indicator with the same controls as the main specification.

After controlling for contractor, year, and environmental covariates, contract-area records stop on average 0.72 taxonomic levels earlier than non-contract-area records. The distribution in Table A5 is consistent with the main result in Table A2. It also shows that the gap is not limited to whether a record reaches species:

Table A5: Distribution of taxonomic stopping levels by zone (% within zone). Each record is assigned the deepest taxonomic level it reaches (0 = no kingdom reported, 7 = species reported). Contract-area records concentrate around class and genus; non-contract records concentrate at species.

Stopping level	Contract	Non-contract
0 — None reported	6.0	0.1
1 — Kingdom	1.0	0.0
2 — Phylum	4.4	2.1
3 — Class	31.4	14.5
4 — Order	10.8	19.3
5 — Family	6.0	4.1
6 — Genus	24.6	12.5
7 — Species	15.8	47.5

Table A6: Linear regression of stopping level on contract-area indicator with the standard control set. The coefficient is interpretable as the average difference in taxonomic depth (out of 7 possible levels) between contract-area and non-contract-area records, conditional on contractor, year, and environmental covariates. Robust (HC1) standard errors in parentheses.

	Stopping level
Contract Area	−0.717***
	(0.023)
Contractor FE	Yes
Year FE	Yes
Environmental Controls	Yes
Num.Obs.	147 341
R2	0.275

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

contract-area records tend to stop earlier in the taxonomic hierarchy more generally.

Identifier Recording by Contractor

Table A7 reports, for each contractor with at least 500 own-tract records, the share of those records that leave the identifier-name field empty. The share ranges from 0% to 100% across contractors.

Administrative Effort by Zone

The difference in field-collection effort across zones noted in Section 5 also appears in the metadata fields that document how each specimen was identified. Table A8 reports completion rates for four of these fields. The first, `taxonomist_yes_no`, records whether a taxonomist was involved in identifying the specimen. This field is completed for 72% of contract-area records, compared with 1% of non-contract-area records. The `identification_method`, `identification_status`, and `identification_date` fields show the same pattern, with completion rates between 68% and 73% in contract areas and between 5% and 6% in non-

Table A7: Share of own-tract specimen records that leave the identifier-name field empty, by contractor. The table includes contractors with at least 500 own-tract records in the analysis sample.

Contractor	Records	Identifier field empty
China Ocean Mineral Resources R...	727	0.0%
Nauru Ocean Resources Inc. - PMN	95,197	0.0%
Institut français de recherche ...	3,913	0.2%
Global Sea Mineral Resources NV...	2,510	2.2%
UK Seabed Resources Ltd. - I - PMN	19,726	15.4%
Federal Institute for Geoscienc...	5,776	26.9%
Ocean Mineral Singapore Pte. Lt...	3,105	27.6%
Interoceanmetal Joint Organizat...	592	79.6%
Government of the Republic of K...	4,097	96.5%
Deep Ocean Resources Developmen...	825	100.0%
Yuzhmorgeologiya - PMN	1,175	100.0%

contract areas. This is consistent with contractors investing substantially more administrative effort in documenting the records that bear on their own permits.[18]

Table A8: Field-completion rates for four compliance-metadata fields by zone. Each cell reports the percentage of records in that zone with a non-missing, non-sentinel value in the named field. Contract-area records are filled in at substantially higher rates across all four fields.

zone	Records	taxonomist_yes_no	identification_method	identification_status	identification_date
APEI	358	0.0%	27.7%	7.8%	7.8%
Contract	151,062	72.2%	67.6%	72.5%	72.5%
Outside	9,547	0.9%	5.5%	4.8%	4.8%

[18] An alternative explanation is that contractors submit records from their own tracts through the ISA-mandated reporting template, which includes these metadata fields, while samples from outside the contractor's tract may not be processed through the same template and may therefore reach the database without those fields populated. In that case, the difference would reflect which submission process a sample passes through rather than a choice about effort. However, the two explanations are not mutually exclusive. Contractors choose how thoroughly to complete mandatory fields, and whether to submit non-contract samples through the template at all is itself a choice about effort. Therefore, this pattern should be thought of as context only.

Extra Figures

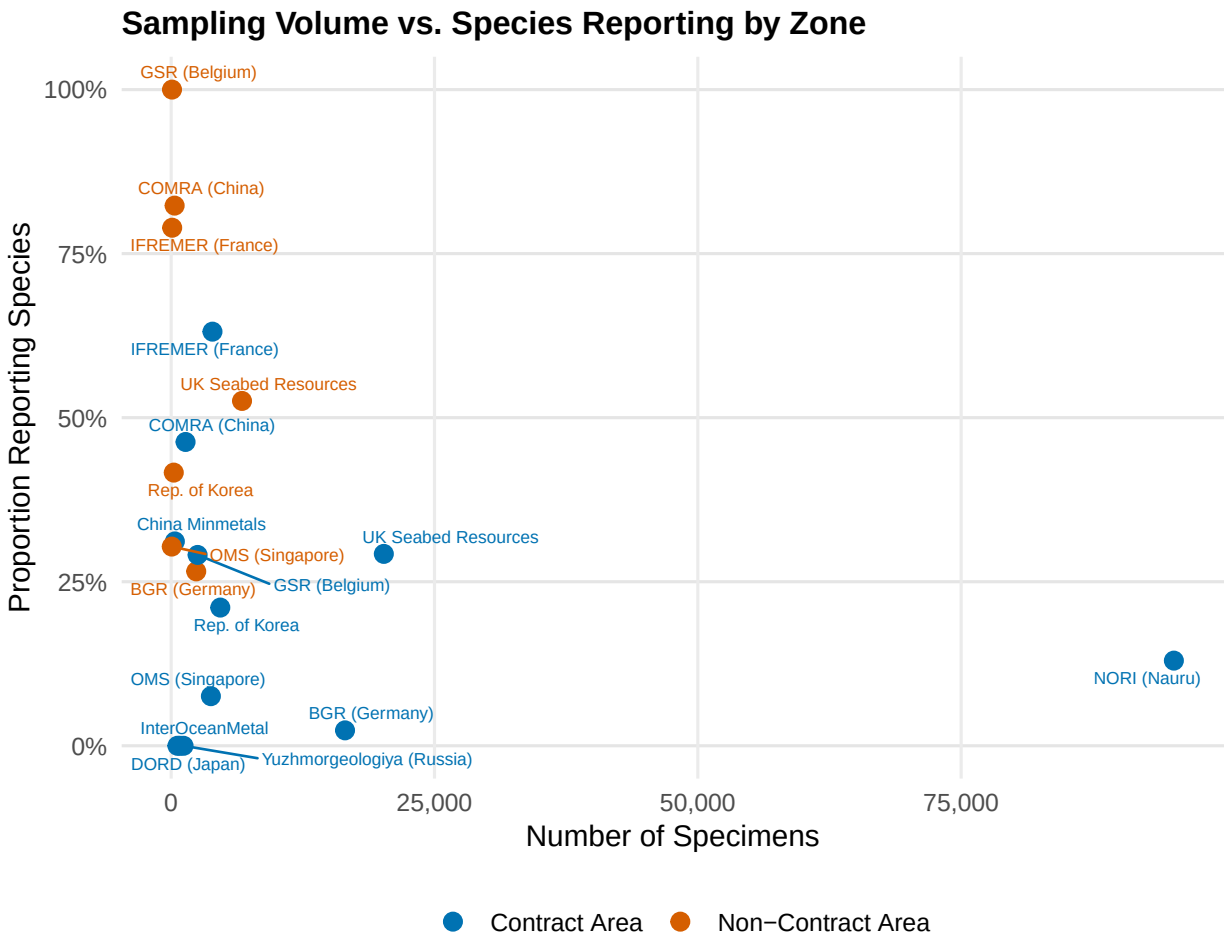


Figure A1: Sampling volume (number of biological specimens) versus species reporting rate by contractor and zone type. Each point represents one contractor's submissions from either contract areas (blue) or non-contract areas (vermillion). Contract areas have more samples but lower species reporting, which is consistent with the model's two predictions and difficult to reconcile with an explanation based on resource constraints.

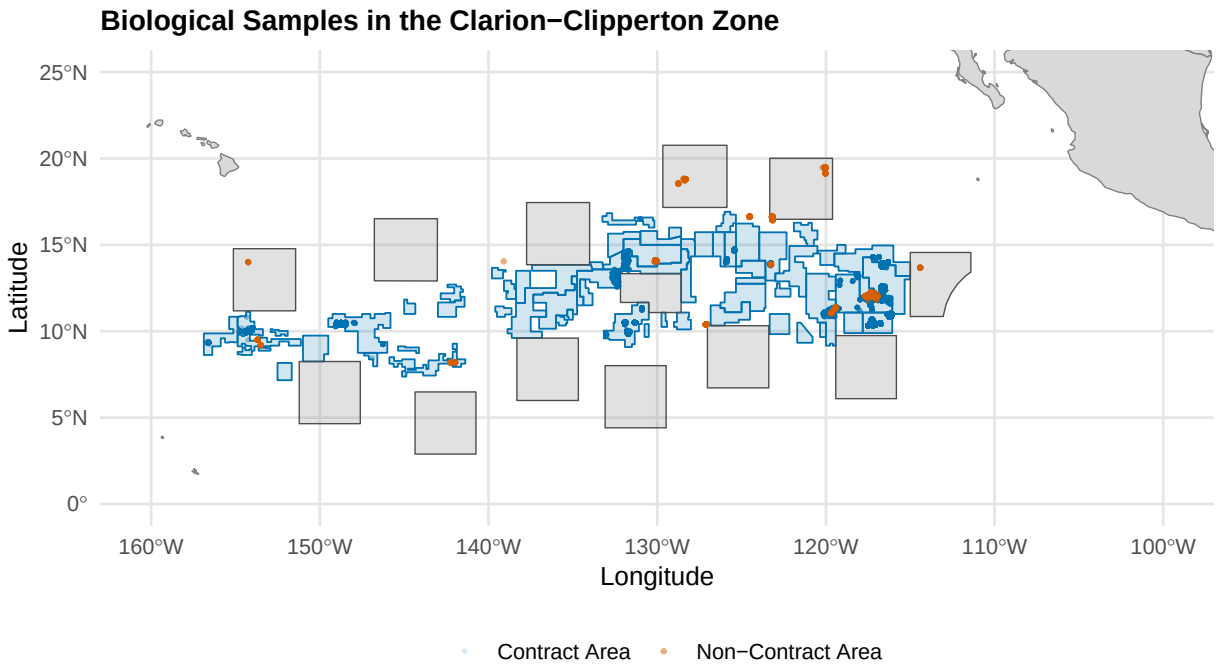


Figure A2: Sample locations overlaid on official ISA contract area polygons (blue) and APEIs (grey). Each point is one biological sample submitted to DeepData. Source for polygons: International Seabed Authority.

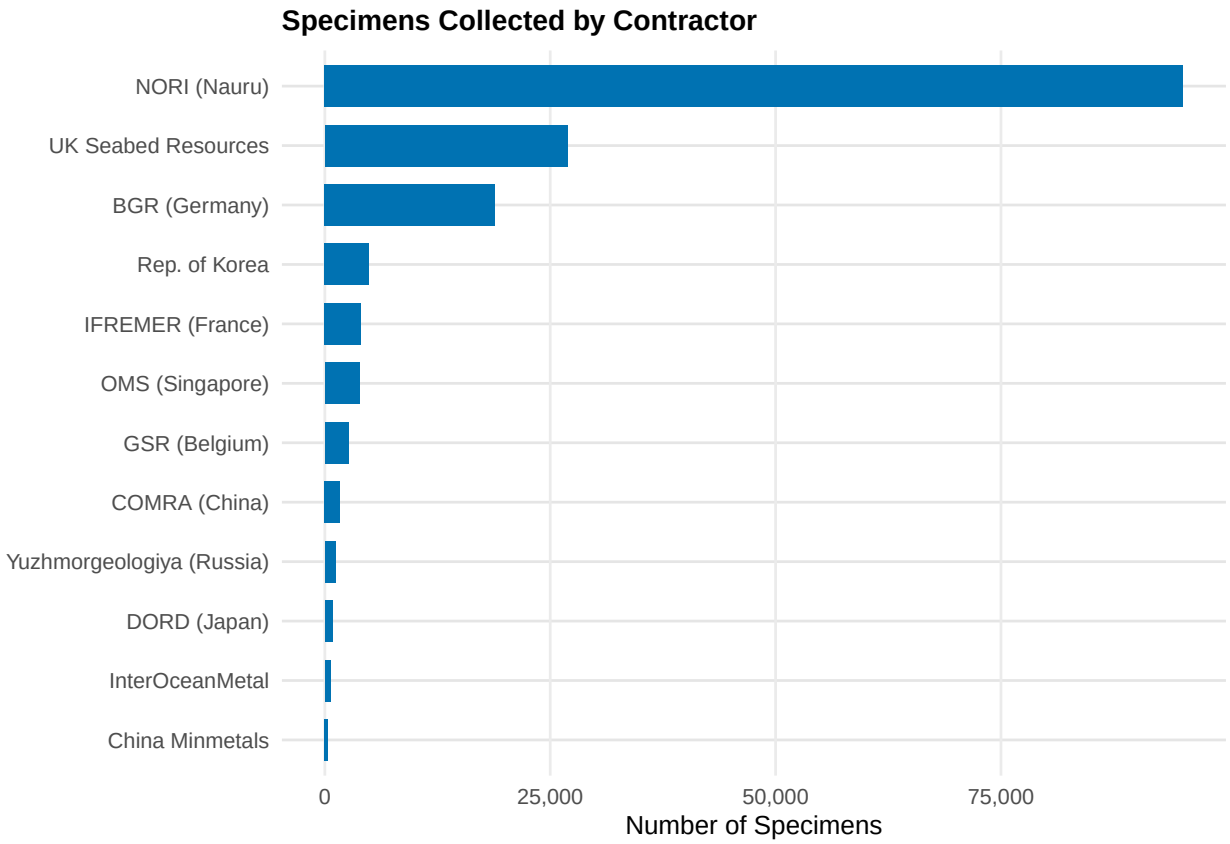


Figure A3: Number of biological specimens collected by each contractor in the CCZ.

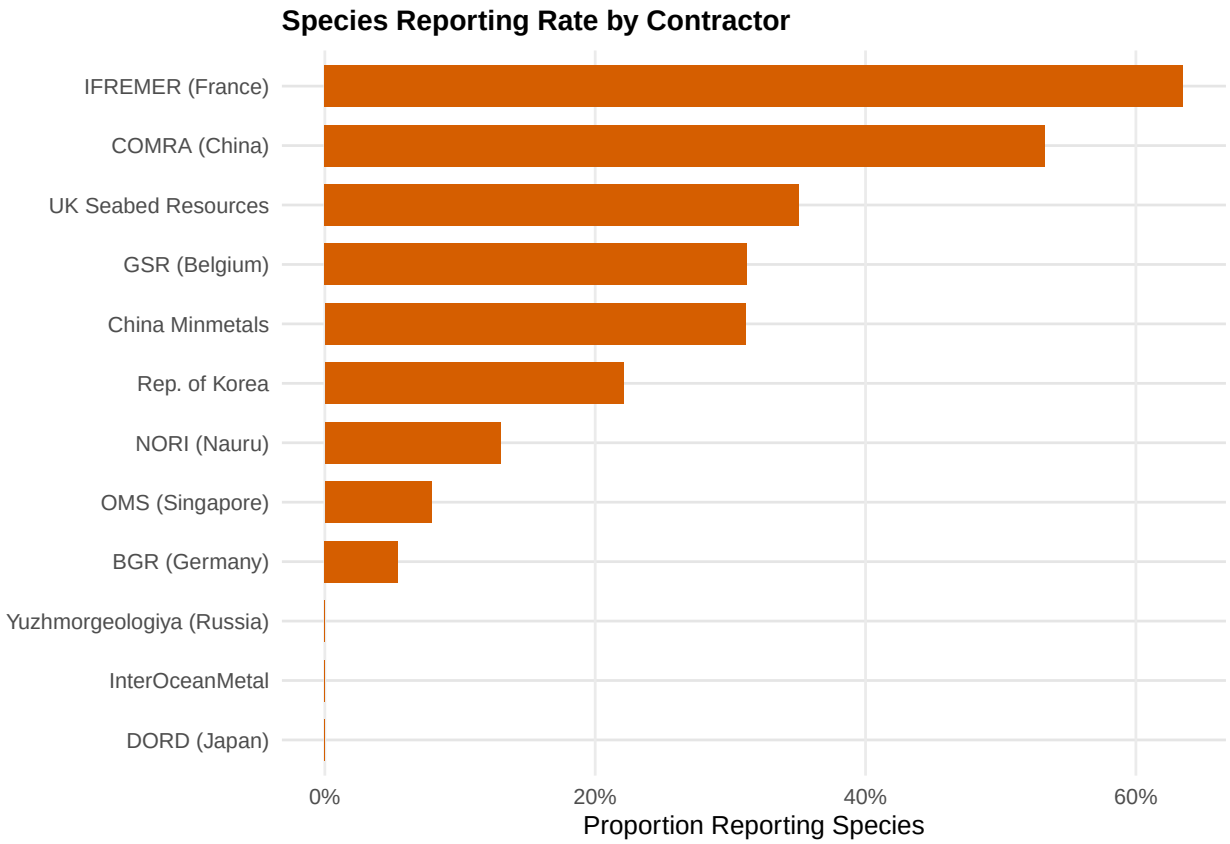


Figure A4: Proportion of observations reporting species-level identification by contractor.

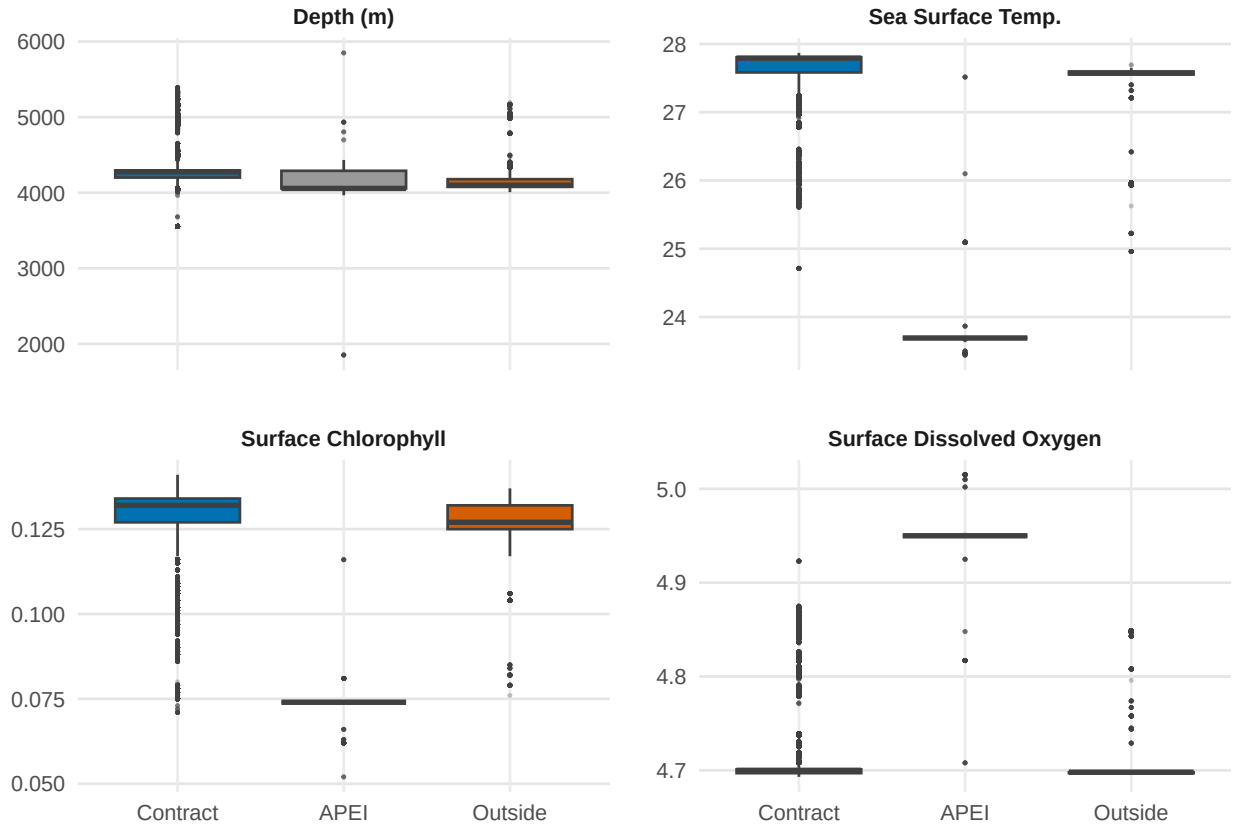


Figure A5: Distributions of the four environmental covariates used in the main analysis, by zone type. Bio-ORACLE values are matched to each sample's longitude and latitude. The contract-area and outside-area distributions overlap heavily. APEI records are similar to contract-area records in depth, but they come from locations with lower sea surface temperature, lower surface chlorophyll, and higher surface dissolved oxygen.

Additional Analyses

Benthic Environmental Controls

The main specification controls for sea surface temperature, surface chlorophyll, surface dissolved oxygen, and depth. Three of these are surface fields, which I use because surface productivity governs the flux of particulate organic carbon that structures abyssal benthic communities. However, a natural question is whether the results change when the controls instead measure conditions at the seafloor. Table A9 re-estimates the main specification with near-bottom temperature, near-bottom dissolved oxygen, and near-bottom current velocity from Bio-ORACLE's bottom-mean layers (Assis et al. 2018) in place of the surface fields, and then with both sets together.

Table A9: Surface versus benthic environmental controls. All columns start from the main specification (column 4 of Table A2) and are estimated on the common sample of records with complete surface and benthic covariates, so that the columns are directly comparable. Surface controls are sea surface temperature, surface chlorophyll, and surface dissolved oxygen; benthic controls are near-bottom temperature, near-bottom dissolved oxygen, and near-bottom current velocity. Depth and contractor and year fixed effects are included in every column. Robust (HC1) standard errors in parentheses.

	Surface (main)	Benthic	Both	Depth only
Contract Area	−0.113***	−0.159***	−0.196***	−0.088***
	(0.006)	(0.007)	(0.007)	(0.006)
Contractor FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Depth	Yes	Yes	Yes	Yes
Surface controls	Yes	No	Yes	No
Benthic controls	No	Yes	Yes	No
Num.Obs.	147 341	147 341	147 341	147 341
R2	0.217	0.223	0.226	0.214

* p < 0.1, ** p < 0.05, *** p < 0.01

The contract-area coefficient is −0.113 with surface controls, −0.159 with benthic controls, and −0.196 with both sets. The gap is negative and significant at $p < 0.001$ in all three columns. That is, neither the sign nor the significance of the result depends on whether the controls are measured at the surface or at the seafloor. However, the magnitude does: across the three sets of controls, the largest estimate is roughly 1.7 times the smallest. The fourth column drops the Bio-ORACLE fields and keeps only depth and the contractor and year fixed effects. The gap in that column is −0.088 ($p < 0.001$), which is smaller than in the other three columns. In other words, adding the Bio-ORACLE fields to a specification that already controls for depth widens the estimated gap, whether those fields are measured at the surface or at the seafloor. I report the surface specification as the main one for the reasons given in Section 4.

It should be noted, however, that the range across these columns implies more uncertainty about the size of the gap than the standard errors alone suggest. Part of the explanation for this sensitivity is that most of these variables vary little within the CCZ. In the analysis sample, the coefficient of variation is 1.7% for sea surface temperature, 7.6% for surface chlorophyll, and 0.8% for surface dissolved oxygen, compared with 3.5% for near-bottom temperature, 3.6% for near-bottom oxygen, and 36.8% for near-bottom current velocity. Near-bottom current velocity is the only one of the six that varies substantially. For comparison, the coefficient of variation of depth is 5.7%.

Station-Level Estimates

The main analysis treats each specimen record as a separate observation. However, records are collected in batches. A sampling station is a single location visited on a single cruise, and it typically corresponds

to one deployment of a box corer, an epibenthic sled, or a remotely operated vehicle. Records from a station almost always share the same contract status, and records from the same station also tend to share the same outcome: in 80% of stations, either every record or no record is identified to species. The data can therefore be interpreted as containing fewer independent observations of contractor behavior than the number of records would suggest. Station sizes also vary enormously. The median station contributes 15 records, while the ten largest together account for 36% of all records, so the record-level estimates give a small number of stations a great deal of weight.

To address both issues, I collapse the data to one observation per station and use the share of the station's records identified to species as the outcome.[19] This outcome corresponds closely to τ in the model, which is defined as the share of recovered organisms that the contractor carries to species level in a given survey. I then re-estimate the main specifications, weighting each station equally and using heteroskedasticity-robust standard errors. The resulting sample contains 1,019 station observations, 119 of them in non-contract areas.

Table A10: Station-level estimates. Each observation is a sampling station (split by contract status where a station spans both); the outcome is the share of the station's specimen records identified to species, and each station is weighted equally. Columns (1)–(4) regress the outcome on the contract-area indicator, adding controls as in the main specifications. Columns (5) and (6) restrict to stations inside some contract polygon and regress the outcome on an indicator for whether the station was sampled by the polygon's owner, with sampler and polygon fixed effects respectively. Heteroskedasticity-robust (HC1) standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

	Main specifications				Within-sampler	Within-polygon
	(1)	(2)	(3)	(4)	(5)	(6)
Contract area / owner	-0.100***	-0.122***	-0.131***	-0.113***	-0.083***	-0.067
	(0.034)	(0.027)	(0.027)	(0.028)	(0.029)	(0.043)
Stations	1,019	1,019	1,019	1,019	990	990
Contractor FE	No	Yes	Yes	Yes	No	No
Year FE	No	No	Yes	Yes	Yes	Yes
Environmental controls	No	No	No	Yes	Yes	Yes
Sampler FE	No	No	No	No	Yes	No
Polygon FE	No	No	No	No	No	Yes

Table A10 reports the results. Without controls, stations in contract areas identify 10 percentage points fewer of their specimens to species than stations elsewhere ($p = 0.003$). Once contractor fixed effects are included, the gap is -0.122 ($p < 0.001$), and it is -0.131 and -0.113 when year fixed effects and environmental controls are added ($p < 0.001$ in the full specification). That is, a typical station in a contract area identifies roughly 11 to 13 percentage points fewer of its specimens to species than a comparable station sampled by the same contractor elsewhere. The within-sampler design gives a similar estimate of -0.083 ($p = 0.004$). The within-

[19] A station is identified by its cruise leg and DeepData station number. The 10 stations whose records span both contract statuses are split into two observations, one for each status, so that contract status is constant within each observation.

polygon design gives −0.067, which is of the same sign and similar size but less precisely estimated ($p = 0.120$), since only 97 stations inside contract polygons were sampled by a guest.

Robustness to Sample Volume

Contract areas are sampled more intensively than non-contract areas (Figure A1). A natural concern is therefore that the contract-area coefficient reflects *sampling volume* rather than a reporting incentive. That is, if operations that process more specimens identify each one less precisely, volume alone could generate the gap. To address this concern, Table A11 adds three measures of volume to the full specification (column 4 of Table A2): the logged number of records that share each observation's *location* (its latitude and longitude), its *research cruise*, and its *identification institution* (the laboratory credited with the identification).

Table A11: Robustness to sample volume. Linear probability models adding the logged number of records at each observation's location, research cruise, and identification institution to the full specification. Dependent variable: indicator for species-level identification. The two institution-volume columns are restricted to records that carry an identifier institution. Robust standard errors in parentheses.

	Baseline	+ Location vol.	+ Cruise vol.	Baseline (ID lab)	+ Institution vol.
Contract area	−0.113***	−0.090***	−0.116***	−0.002	0.020***
	(0.006)	(0.006)	(0.007)	(0.007)	(0.007)
log(Location volume)		0.063***			
		(0.000)			
log(Cruise volume)			−0.027***		
			(0.003)		
log(Institution volume)					0.116***
					(0.004)
Contractor FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes	Yes
Num.Obs.	147 341	147 341	147 341	126 099	126 099
R2	0.217	0.323	0.217	0.308	0.325

* p < 0.1, ** p < 0.05, *** p < 0.01

The location and cruise controls (columns 1–3) are estimated on the full sample. The contract-area coefficient is −0.113 in the baseline, −0.090 with the location control, and −0.116 with the cruise control, so the gap remains in each case. The volume terms themselves do not all have the same sign. Records from more heavily sampled *locations*, and records handled by *institutions* that process more specimens, are reported with *higher* precision. Records from larger *cruises* are reported with somewhat lower precision. That is, the cruise is the only level at which higher volume is associated with lower precision.

The institution-volume control (columns 4–5) can be computed only for the 86% of records that carry an identifier institution. On that subsample, the contract-area coefficient is −0.002 before any volume control is added (column 4) and +0.020 with it (column 5). That is, there is no gap on this subsample for volume

to explain, which reflects which records name an institution, as discussed in Section 5.6. Sample volume, at every level at which I can measure it, therefore does not account for the contract-area gap.